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# Likelihood and pivotal-based reliability inference for inverse exponentiated Rayleigh distribution under block progressive censoring

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**ABSTRACT**

In this paper, focus is on the estimation of survival characteristics, specifically the reliability function, hazard rate function, median time to failure, and differences in different test facilities, using block progressive censored data. Estimations are carried out through both maximum likelihood and pivotal methods, assuming the lifetime distribution of test units follows an inverse exponentiated Rayleigh distribution. The paper derives maximum likelihood estimators for unknown parameters, exploring their existence and uniqueness properties. Approximate confidence intervals for survival characteristics are constructed using the delta method and likelihood theory. Moreover, point and generalized confidence interval estimators are developed through a pivotal quantity-based method. A simulation study is conducted to compare the performance of the proposed approaches, revealing that the pivotal quantity-based approach yields superior estimation results. Finally, the proposed estimates are applied to analyze two real datasets, illustrating their practical applicability.

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Block censoring; difference in different test facilities; inverted exponentiated Rayleigh distribution; pivotal inference; progressive Type-II censoring scheme; survival characteristics

## 1. Introduction

In the lifetime data analysis, the practitioners come across with numerous situations in which the test subjects are either not permitted to participate in the study or left the study willingly before their failure occurred. There could be so many facts in this regard, but budget and time constraints may be crucial factors. In order to deal with or overcome these situations, life test study of subjects are either terminated after pre-fixed time point or terminated after a pre-fixed number of subject failures are observed. Such termination of subjects under study is commonly known as censoring, the first termination case is known as Type-I censoring and the latter is known as Type-II censoring. One shortcoming that both of these censoring techniques have in common is that live test subjects cannot be pulled out of the study somewhere in the middle which may be the more practical case in many situations. For instance, a subscriber may cancel their subscription in the middle of the study according to an OTT platform's study of customer subscription life length. In some cases, an electrical or mechanical component may break accidentally before failing.

In the literature, progressive Type-II censoring overcomes these shortcomings which allows live test subjects to be removed at stages other than the eventual termination point. Therefore, it is more adaptable and efficient than conventional censoring techniques. The process of progressive

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censoring technique is explained as follows. Assume $n$ identical items are put on a life-testing experiment. It is decided that only $m$ failures will be observed and the remaining $n - m$ items will then be progressively censored as follows. At the first failure, remove $R_1$ of the surviving $n - 1$ items from the experiment. At the next failure, remove $R_2$ of the remaining $n - 2 - R_1$ surviving items from the experiment. Repeat this censoring procedure up to the $m$th failure, then we have all remaining $n - m - \sum_{i=1}^{m-1} R_i$ surviving items. Finally remove all these remaining surviving items from the experiment at the $m$th failure. For more details on this censoring, one could also refer to the monographs (Balakrishnan & Aggarwala, 2000; Balakrishnan & Cramer, 2014) for a comprehensive review. Among others, one may refer to (Abdel-Hamid & Al-Hussaini, 2011; Balakrishnan & Sandhu, 1995; Balakrishnan et al., 2001; Burkschat et al., 2006; Balakrishnan & Han, 2007; Bhattacharya et al., 2016; Bhattacharya & Balakrishnan, 2022; Chakraborty et al., 2023; Dey & Dey, 2014; Kayal et al., 2017; Kohansal et al., 2022; K. Singh et al., 2023; Kundu, 2008; Krishna & Kumar, 2013; Lodhi, Tripathi, & Rastogi, 2021; Lin et al., 2004; Maurya et al., 2019; Mahto & Tripathi, 2020; Ng et al., 2002; Pradhan & Kundu, 2009; Raqab et al., 2010; Rastogi et al., 2012; S. Singh et al., 2015; Tse & Yang, 2003; Wu et al., 2005; Wang, 2018) and the references cited therein.

Sometimes, for one person, checking each subject under study simultaneously is not usually a practical option. For example, the equipment needed to conduct the life test on all of the units at once may not easily accessible. To address this drawback, a block censoring mechanism was recently introduced by (Ahmadi et al., 2018). According to this mechanism, a total of $n$ units are stochastically divided into $k$ sets where the $i$th set, $i = 1, 2, \cdots, k$, contains $n_i$ units such that $\sum_{i=1}^{k} n_i = n$ and then these $k$ groups may be tested with $k$ facilities under complete or censored mechanisms. (Ahmadi et al., 2018) made an excellent discussion on the block censoring techniques and considered the case of testing $k$ groups with $k$ facilities under Type-II censoring when the units life follows an exponential distribution. Also, (Zhu, 2020) studied the Weibull distribution under block Type-II censoring using maximum likelihood and Bayesian methods. In the area of life testing, modern products often boast high reliability and long lifetimes, rendering the acquisition of complete failures for all test units impractical. Consequently, datasets gathered frequently exhibit insufficient samples or small sample sizes, leading to less accurate estimations when employing the maximum likelihood method. The variance of estimators tends to increase as the sample size decreases, further complicating the efficiency of results. Moreover, interval estimates derived through likelihood theory offer only approximate confidence intervals, relying on large sample theory and requires a derivative process of the likelihood function. While Bayesian methods address these challenges, they come with their own set of drawbacks. Notably, the requirement to treat all model parameters, including nuisance parameters, as random variables poses a challenge. In other words, Bayesian methods demand prior information on unknown parameters, introducing subjectivity into the analysis. Furthermore, these methods can be cumbersome and computationally expensive, adding an additional layer of complexity to their application. In the view of this unsatisfactory state of estimation methods, (Weerahandi, 1993, 2004, 2012) introduced the pivotal quantity-based generalized point and confidence interval estimates where the other classical and Bayesian approach fail or yield unreliable results. Pivotal quantity-based method requires only some mathematical transformations and may be the better option. Further, studying the case of testing $k$ groups with $k$ facilities under progressive Type-II censoring may improve efficiency and save the cost and time. Therefore, in this work, our main motive and contribution is to draw pivotal quantity-based inference under block progressive censored data.

Inverse exponentiated Rayleigh (IER) distribution is a special case in the family of inverse exponentiated distributions. (Maurya et al., 2019) conducted a visual examination of the hazard rate function associated with this particular distribution. Their findings revealed hazard rate pattern that initially starts at a lower value, then increases up to a specific time point, and subsequently decreases. This non-monotonic nature implies suitability of the distribution for modeling phenomena characterized by early failure behavior. Examples of such phenomena include mechanical or

electrical device failures, infant mortality, fatigue failure, capacity and voltage degradation in batteries, potential-induced degradation in solar panels, sensor malfunctions in medical devices, and patterns in patient treatment. Studies of various degradation phenomena often lead to such hazard rate function. However, it lacks fitting bathtub-shapes of hazard rate function. Despite this limitation, (Maurya et al., 2019) identified it as a superior alternative to other well-known distributions, such as Gompertz, inverse exponential, Burr-XII, and the Generalized Rayleigh distribution. Notably, (Ghitany et al., 2014) and (Rastogi & Tripathi, 2014) independently affirmed its superiority over other distributions within its family. Furthermore, the distribution demonstrated its efficacy when compared to well-known statistical distributions. For instance, the hazard rate exhibited similar behavior to distributions like lognormal, inverse Weibull, and generalized inverted exponential, as noted by (Panahi & Moradi, 2020). The family of this distribution was first proposed by (Ghitany et al., 2014) and investigated many of their useful structural characteristics. Since then, a number of scholars have examined this distribution in other contexts; for examples, see (Rastogi & Tripathi, 2014) investigated various cases of maximum likelihood and Bayes estimators for this distribution under progressive censoring. (Sriramachandran & Palanivel, 2014) studied acceptance sampling plans. (Kohansal, 2017) studied large estimation of the stress-strength reliability under progressive censored data. (Kayal et al., 2018) studied estimation and prediction under hybrid censoring. (Panahi & Asadi, 2019) studied the estimation of stress-strength using micro-splat splashing data. Further, one may refer to (Dutta et al., 2023; Kumari, Tripathi, et al., 2022; Kizilaslan, 2018; Kumari, Lodhi, et al., 2022; Lodhi, Tripathi, & Wang, 2021; Maurya et al., 2019; Mahto & Tripathi, 2020; Wang et al., 2020) and the references cited therein for some further discussion on the applicability of this family of densities in real-life situations. Motivated by its popularity, we consider the problem of deriving inferences using IER distribution under block censored data.

Reliability engineering practitioners may benefit from using information derived from survival characteristics (SCs) of products or units, such as reliability, hazard rate, and mean or median time to failure, to fairly decide whether to overhaul or replace the component. Consequently, there is a keen interest to derive inferences on these SCs when the lifetime of units follows IER distribution under block progressive censored data. On a related note, block censored mechanism frequently distributes the test units to various test facilities, and the test records of the test units afterwards come from these test facilities. Most professionals are aware that even when two seemingly identical testing facilities are used to test the same units, the test records from those facilities are frequently not exactly the same. The test disparities can sometimes be rather large. Deriving inferences on differences in different test facility (DDTF) is, therefore, another aspect of our work. To address these two concerns, we proposed an algorithm to derive pivotal quantity-based inferences and compare its outcomes with that of the likelihood-based inference.

The main body of the article is presented here after. In Section 2, we formulate the issue and stipulate the necessary assumptions. Using the maximum likelihood method, we generate the SC and DDTF inferences, and the observed data and the delta method are used to provide the related confidence intervals. In Section 3, we provide a pivotal-based inferences on the SC and DDTF and propose an algorithm in this regard. We provide detailed simulation experiments in Section 4 for the purpose of assessing the proposed strategies. In Section 5, there are two examples with illustrations. Finally, Section 6 offers a few conclusions.

## 2. Problem formulation and maximum likelihood estimation

In this section, we provide the problem formulation and necessary assumptions, and use maximum likelihood method to infer SC and the DDTF.

### 2.1. Problem formulation and assumptions

Suppose $n$ identical units are independently put on a life test based on block progressive censoring mechanism. Also, suppose that the lifetimes of units are following an IER distribution whose probability density function, say $f(t;\theta,\lambda)$ is given by

$$f(t;\theta,\lambda) = 2\theta\lambda t^{-3} e^{-\lambda t^{-2}} \left(1 - e^{-\lambda t^{-2}}\right)^{\theta-1}, t>0 \tag{1}$$

The corresponding cumulative distribution function $\mathcal{F}(t;\alpha,\lambda)$, reliability function $\mathcal{R}(t;\theta,\lambda)$ and hazard rate function $h(x;\theta,\lambda)$ are, respectively, given by

$$\mathcal{F}(t;\theta,\lambda) = 1 - \left(1 - e^{-\lambda t^{-2}}\right)^{\theta}, t>0 \tag{2}$$

$$\mathcal{R}(t;\theta,\lambda) = \left(1 - e^{-\lambda t^{-2}}\right)^{\theta}, t>0 \tag{3}$$

and

$$h(t;\theta,\lambda) = 2\theta\lambda t^{-3} e^{-\lambda t^{-2}} \left(1 - e^{-\lambda t^{-2}}\right)^{-1} \tag{4}$$

It is noted that $\theta>0$ and $\lambda>0$ respectively denote shape and scale parameters. Mean time to failure is an important survival characteristic on the lifetime of the population of interest, but its expression for this distribution is not in closed form. Therefore, a similar kind of survival characteristic may be the median time to failure. The median time to failure (MdTF) for this distribution is defined as follows:

$$\text{MdTF} = \sqrt{\frac{-\lambda}{\log(1-(0.5)^{1/\theta})}} \tag{5}$$

As discussed by (Zhu, 2020), by assuming a single parameter, we can account for the DDTF. We can statistically account for the DDTF by defining $\mathcal{T}_{ij}$ as the random variable corresponding to the $j^{\text{th}}$ observed test record of the $i^{\text{th}}$ test facility as follows:

$$\mathcal{T}_{ij} \sim \text{IER}(\theta_i,\lambda), i = 1,2,3,\ldots,k; j = 1,2,\ldots,n_i. \tag{6}$$

Moreover, as many test facilities may share similar intrinsic failure causes, we presumptuously used the common shape parameter in those facilities. Scale parameter $\theta_i$, may also be called as the $i^{\text{th}}$ test facility effect, describes the difference in test results for the same unit from different test facilities. The differences in $\theta_i, i = 1,2,\ldots,k$, are what make up the DDTF, hence estimating these values is crucial to infer DDTF. In our work, the DDTF is estimated using the difference between $\theta_i$ interval estimations, $i = 1,2,\cdots,k$. If the DDTF is non-negligible, $\theta$ in Eq. (1) can be calculated using a weighted estimation of $\theta_i$, then $\theta = \sum_{i=1}^{m} \omega_i\theta_i / \sum_{i=1}^{m} w_i$ and $\omega_i$ is the weight coefficient.

### 2.2. Maximum likelihood estimation (MLE)

In this section, utilising the approach of maximum likelihood, we derived the estimates of reliability Eq. (3), hazard rate Eq. (4), median time to failure Eq. (5), and the DDTF (i.e. the differences in $\theta_i, i = 1,2,\ldots,k$). This is based on the formulation of the problem and essential assumptions in Section 2.1. In testing procedure, a prefixed $m_i$ number of observed failures and prefixed removals $r_i = (r_{i1}, r_{i2}, \cdots, r_{im_i})$ are given in advance for the $i^{th}$ testing group and a set of progressive failure times $(\mathcal{T}_{i1}, \mathcal{T}_{i2}, \cdots, \mathcal{T}_{im_i})$ is obtained under this $i^{th}$ group. Therefore, one could observe the block progressive censored data as follows:

$$\begin{cases} \textit{Data} & \textit{Group} \\ (\mathcal{T}_{11}, r_{11}), (\mathcal{T}_{12}, r_{12}), \cdots, (\mathcal{T}_{1s_1}, r_{1m_1}) & 1 \\ (\mathcal{T}_{21}, r_{21}), (\mathcal{T}_{22}, r_{22}), \cdots, (\mathcal{T}_{2s_2}, r_{2m_2}) & 2 \\ \cdots & \cdots \\ (\mathcal{T}_{k1}, r_{k1}), (\mathcal{T}_{k2}, r_{k2}), \cdots, (\mathcal{T}_{km_k}, R_{km_k}) & k. \end{cases} \tag{7}$$

Let $\mathcal{T}_i = (\mathcal{T}_{i1}, \mathcal{T}_{i2}, \cdots, \mathcal{T}_{im_i})$ be the progressively censored sample from IER$(\theta_i, \lambda)$ distribution under $i^{th}$ test group, then the likelihood function under $i^{th}$ test group can be written as follows:

$$L(\theta_i, \lambda; \mathcal{T}_i) = C_i \prod_{j=1}^{m_i} f(\mathcal{T}_{ij}, \theta_i, \lambda) \left[1 - \mathcal{F}(\mathcal{T}_{ij}, \theta_i, \lambda)\right]^{r_{ij}}$$

where $C_i = n_i \prod_{j=1}^{m_i-1} (n_i - j - \sum_{l=1}^{j} r_{il})$ is a normalizing constant, and $f(.)$ and $\mathcal{F}(.)$ are the probability density and cumulative distribution function in Eq. (1) and Eq. (2). Further, the likelihood function under the block progressive censored data in Eq. (7) for $\theta = (\theta_1, \theta_2, \cdots, \theta_k)$ and $\mathcal{T} = (\mathcal{T}_1, \mathcal{T}_2, \cdots, \mathcal{T}_k)$ can be expressed as follows:

$$L(\theta, \lambda; \mathcal{T}) = \prod_{i=1}^{k} C_i \prod_{j=1}^{m_i} f(\mathcal{T}_{ij}, \theta_i, \lambda) \left[1 - \mathcal{F}(\mathcal{T}_{ij}, \theta_i, \lambda)\right]^{r_{ij}}$$

Ignoring the constant terms, the likelihood function can be obtained as given in the following expression

$$L(\theta, \lambda; \mathcal{T}) \propto \left(\prod_{i=1}^{k} \theta_i^{m_i}\right) \left(\lambda^{\sum_{i=1}^{k} m_i}\right) \left(\prod_{i=1}^{k} \prod_{j=1}^{m_i} \mathcal{T}_{ij}^{-3}\right) \left(\prod_{i=1}^{k} \prod_{j=1}^{m_i} \left(\frac{e^{-\lambda \mathcal{T}_{ij}^{-2}}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}\right)\right) \times e^{\sum_{i=1}^{k} \sum_{j=1}^{m_i} \theta_i (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}$$

The associated log-likelihood function, denoted as $l(\theta, \lambda; \mathcal{T})$, can be obtained as

$$\begin{aligned} l(\theta, \lambda; \mathcal{T}) \quad &\propto \sum_{i=1}^{k} m_i \log(\theta_i) + \log(\lambda) \sum_{i=1}^{k} m_i + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \log\left(\frac{e^{\lambda \mathcal{T}_{ij}^{-2}}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}\right) \\ &+ \sum_{i=1}^{k} \theta_i \sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right) \end{aligned} \tag{8}$$

The maximum likelihood estimates of the unknown parameters can be obtained by maximizing the log-likelihood function in Eq. (8). Before we proceed with the maximum likelihood estimation of the parameters $\theta$ and $\lambda$, we first present some results as below.

**Theorem 2.1.** For a given $\lambda$ and $m_i > 0$, $i = 1, 2, \cdots, k$, the maximum likelihood estimator of $\theta_i$, denoted as $\hat{\theta}_i$, can be obtained as given below:

$$\hat{\theta}_i = \frac{m_i}{-\sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)} \tag{9}$$

*Proof.* Differentiating Eq. (8) with respect to $\theta_i$, we can have the following equality

$$\hat{\theta}_i = \frac{m_i}{-\sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}$$

Also, we can have the inequality $\log\left(\frac{\theta_i}{\hat{\theta}_i}\right) \leq \frac{\theta_i}{\hat{\theta}_i} - 1$ for $\frac{\theta_i}{\hat{\theta}_i} > 0$ i.e.

$$\log(\theta_i) \leq \frac{-\theta_i \sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}{m_i} - 1 + \log\left(\hat{\theta}_i\right)$$

Ignoring the constant terms in Eq. (8) and applying the above inequality, we can get the following result:

$$\begin{aligned} l(\theta, \lambda; \mathcal{T}) \quad & \leq \sum_{i=1}^{k} m_i \log(\hat{\theta}_i) + \log(\lambda) \sum_{i=1}^{k} m_i + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \log\left(\frac{e^{\lambda \mathcal{T}_{ij}^{-2}}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}\right) \\ & \sum_{i=1}^{k} \theta_i \sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right) \end{aligned}$$

Now, using the result of $\hat{\theta}_i$, i.e. $m_i = -\hat{\theta}_i \sum_{j=1}^{m_i} (1 + r_{ij}) \log(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}})$, the above inequality may be further deduced as follows

$$\begin{aligned} l(\theta, \lambda; \mathcal{T}) \quad & \leq \sum_{i=1}^{k} m_i \log(\hat{\theta}_i) + \log(\lambda) \sum_{i=1}^{k} m_i + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \log\left(\frac{e^{\lambda \mathcal{T}_{ij}^{-2}}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}\right) \\ & \sum_{i=1}^{k} \hat{\theta}_i \sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right), \\ l(\theta, \lambda; \mathcal{T}) \quad & \leq l(\hat{\theta}, \lambda; \mathcal{T}) \end{aligned}$$

The equality will hold if and only if $\theta = \hat{\theta} = \left(\hat{\theta}_1, \hat{\theta}_2, \cdots, \hat{\theta}_k\right)$. Hence, the theorem is proved.

Using Theorem 2.1 and ignoring the constant terms, we put the value of $\hat{\theta}_i$ in Eq. (8), then we get the profile log likelihood as given below:

$$\begin{aligned} l(\lambda) \quad & \propto \log(\lambda) \sum_{i=1}^{k} m_i + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \log\left(\frac{e^{\lambda \mathcal{T}_{ij}^{-2}}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}\right) \\ & - \sum_{i=1}^{k} m_i \log\left(-\sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)\right) \end{aligned} \tag{10}$$

We obtain the following result to find the maximum likelihood estimator of the $\lambda$.

**Theorem 2.2.** For $m_i > 0$, $i = 1, 2, \cdots, k$, the maximum likelihood estimator of $\lambda$, denoted as $\hat{\lambda}$, exists and can be uniquely obtained by solving the following non-linear equation

$$\Phi(\lambda) = 0, \tag{11}$$

*where* $\Phi(\lambda) = \frac{\sum_{i=1}^{k} m_i}{\lambda} - \sum_{i=1}^{k} \sum_{j=1}^{m_i} \frac{\mathcal{T}_{ij}^{-2}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}} + \sum_{i=1}^{k} m_i \left( \frac{\sum_{j=1}^{m_i} (1 + r_{ij}) \frac{e^{-\lambda \mathcal{T}_{ij}^{-2}} \mathcal{T}_{ij}^{-2}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}}{\sum_{j=1}^{m_i} (1 + r_{ij}) \left| \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right) \right|} \right)$

*Proof.* The likelihood equation for the MLE of $\lambda$ in Eq. (11) can be obtained directly by differentiating $l(\lambda)$ with respect to $\lambda$ and setting it to zero. In the following, the existence and uniqueness of the MLE of $\lambda$ are established.

We can write $\Phi(\lambda) = \sum_{i=1}^{k} \Phi_i(\lambda)$ where

$$\Phi_i(\lambda) = \frac{m_i}{\lambda} - \sum_{j=1}^{m_i} \frac{\mathcal{T}_{ij}^{-2}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}} + m_i \left( \frac{\sum_{j=1}^{m_i} (1 + r_{ij}) \frac{e^{-\lambda \mathcal{T}_{ij}^{-2}} \mathcal{T}_{ij}^{-2}}{1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}}}{\sum_{j=1}^{m_i} (1 + r_{ij}) \left| \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right) \right|} \right)$$

Following (Ghitany et al., 2014) and using the Theorem 1 in their work, it can be easily shown that $\Phi_i(0) = \infty$, $\Phi_i(\infty) < 0$, and $\Phi_i(\lambda)$ is a decreasing function in $\lambda$ for each $i = 1, 2, \cdots, k$.

Consequently, it can be shown that $\Phi(0) = \infty$, $\Phi(\infty) < 0$, and $\Phi(\lambda)$ is a decreasing function in $\lambda$ which implies that the Eq. (11) has a unique solution.

Since, it is not possible to find the solution of Eq. (11) with respect to $\lambda$ in a closed form, any computational numerical iterative method may be adopted to find the solution. Once, we obtain the maximum likelihood estimate of $\lambda$, say $\hat{\lambda}$, the maximum likelihood estimates of $\theta_i$ can be obtained using Theorem 2.1 as

$$\hat{\theta}_i = \frac{-m_i}{\sum_{j=1}^{m_i} (1 + r_{ij}) \log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}$$

Based on the invariance property of the maximum likelihood estimators, substitute the values $(\hat{\theta}, \hat{\lambda})$ in Eq. (3), Eq. (4) and Eq. (5) to find the maximum likelihood estimators of the reliability, hazard rate and median time to failure, denoted as $\hat{\mathcal{R}}(t; \theta, \lambda)$, $\hat{h}(t; \theta, \lambda)$ and $\widehat{MdTF}$, respectively, here

$$\hat{\theta} = \frac{\sum_{i=1}^{k} \hat{w}_i \hat{\theta}_i}{\sum_{i=1}^{k} \hat{w}_i}$$

where $\hat{w}_i = 1/var\left(\hat{\theta}_i\right)$ is the estimator of $w_i$ and $var\left(\hat{\theta}_i\right)$ is the observed variance of parameter $\theta_i$ under $i^{\text{th}}$ testing group provided in the next subsection.

### 2.3. Approximate confidence intervals

We are unable to determine the confidence intervals of the $v = (\theta, \lambda)$ because the maximum likelihood estimators of $v$ cannot be calculated in closed forms, making it difficult to determine the exact distributions of $\hat{v}$. However, in order to estimate the DDTF, we plan to apply the large sample approximation for the confidence intervals of $\theta_i; i = 1, 2, \ldots, k$. It is well known that the asymptotic distribution of the $\hat{v}$ is the normal distribution $N_{k+1}(v, I^{-1}(v))$. In real life, we typically estimate $I^{-1}(v)$ by $I^{-1}(\hat{v})$, where $I^{-1}(\hat{v})$ is the inverse of the observed information matrix of $v$, which is provided below:

$$I^{-1}(\hat{v}) = I^{-1}(\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k, \hat{\lambda}) = -\begin{bmatrix} \frac{\partial^2 l}{\partial \lambda^2} & \frac{\partial^2 l}{\partial \lambda \partial \theta_1} & \frac{\partial^2 l}{\partial \lambda \partial \theta_2} & \cdots & \frac{\partial^2 l}{\partial \lambda \partial \theta_k} \\ \frac{\partial^2 l}{\partial \theta_1 \partial \lambda} & \frac{\partial^2 l}{\partial \theta_1^2} & \frac{\partial^2 l}{\partial \theta_1 \partial \theta_3} & \cdots & \frac{\partial^2 l}{\partial \theta_1 \partial \theta_k} \\ \vdots & \vdots & \vdots & \ddots & \\ \frac{\partial^2 l}{\partial \theta_k \lambda} & \frac{\partial^2 l}{\partial \theta_k \partial \theta_2} & \frac{\partial^2 l}{\partial \theta_k \partial \theta_3} & \cdots & \frac{\partial^2 l}{\partial \theta_k^2} \end{bmatrix}^{-1}_{(\hat{\theta}_1, \ldots, \hat{\theta}_k, \hat{\lambda})}$$

where the associated elements are

$$\frac{\partial^2 l}{\partial \lambda^2} = -\frac{\sum_{i=1}^{k} m_i}{\lambda^2} - \sum_{i=1}^{k} \theta_i \sum_{j=1}^{m_i} (1 + r_{ij}) \frac{\left(\mathcal{T}_{ij}^{-2}\right)^2}{\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)^2} + \sum_{i=1}^{k} \sum_{j=1}^{m_i} \frac{\left(\mathcal{T}_{ij}^{-2}\right)^2}{\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)^2}$$

$$\frac{\partial^2 l}{\partial \theta_i^2} = -\frac{m_i}{\theta_i^2}, \frac{\partial^2 l}{\partial \theta_i \partial \lambda} = \frac{\partial^2 l}{\partial \lambda \partial \theta_i} = \sum_{j=1}^{m_i} (1 + r_{ij}) \frac{e^{-\lambda \mathcal{T}_{ij}^{-2}} (\mathcal{T}_{ij}^{-2})}{\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}, i = 1, 2, 3, \ldots k$$

and the remaining elements will be zero. Therefore, the approximate $100(1 - \tau)\%$ CI of $v$ is

$$\left(\hat{v} - z_{\tau/2} \sqrt{var(\hat{v})}, \hat{v} + z_{\tau/2} \sqrt{var(\hat{v})}\right) \tag{12}$$

where $Z_{\tau/2}$ represents the upper $(\tau/2)^{th}$ percentile of the standard normal distribution. Maximum likelihood inference on the DDTF is interpreted using the differences in the confidence intervals of the $\theta_i (i = 1, 2, \ldots, k)$. Furthermore, to obtain confidence intervals for $\theta$, reliability, hazard rate and MdTF, we need to find their variances. The delta method is a general approach for estimating the variance of $\hat{\mathcal{R}}(t;\theta,\lambda), \hat{h}(t;\theta,\lambda)$ and $\widehat{\text{MdTF}}$. Based on the asymptotic normality of the maximum likelihood estimators, the approximate confidence intervals (ACIs) of $\theta$, reliability, hazard rate and MdTF are obtained. Let $g(\theta,\lambda)$ be an arbitrary function of the parameters $\theta$ and $\lambda$. Then the asymptotic variance of the function $g(\theta,\lambda)$ can be obtained as

$$\widehat{var}(\hat{g}) = [\nabla g^T(\theta,\lambda)] I^{-1}(\hat{\theta},\hat{\lambda}) [\nabla g(\theta,\lambda)] \tag{13}$$

where

$$\nabla g(\theta,\lambda) = \left(\frac{\partial g}{\partial \theta_1}, \frac{\partial g}{\partial \theta_2}, \cdots, \frac{\partial g}{\partial \theta_k}\right)\Bigg|_{(\theta,\lambda)=(\hat{\theta},\hat{\lambda})} \text{for } g(\theta,\lambda) = \theta = \sum_{i=1}^{k} \omega_i \theta_i / \sum_{i=1}^{k} w_i$$

and

$$\nabla g(\theta,\lambda) = \left(\frac{\partial g}{\partial \theta_1}, \frac{\partial g}{\partial \theta_2}, \cdots, \frac{\partial g}{\partial \theta_k}, \frac{\partial g}{\partial \lambda}\right)\Bigg|_{(\theta,\lambda)=(\hat{\theta},\hat{\lambda})}$$

for $g(\theta,\lambda)$ being reliability, hazard rate and MdTF. Therefore, the asymptotic distribution of $g(\theta,\lambda)$ can be constructed as

$$\frac{\hat{g}(\theta,\lambda) - g(\theta,\lambda)}{\sqrt{\widehat{var}(\hat{g})}} \to N(0,1)$$

here $\hat{g}(\theta,\lambda) = g(\hat{\theta},\hat{\lambda})$ denotes the maximum likelihood estimate of $g(\theta,\lambda)$. Thus, the $100(1-\tau)\%$ ACIs of $\hat{g}(\theta,\lambda)$ can be obtained as

$$\left(\hat{g}(\theta,\lambda) - Z_{\tau/2}\sqrt{\widehat{var}(\hat{g})}, \hat{g}(\theta,\lambda) + Z_{\tau/2}\sqrt{\widehat{var}(\hat{g})}\right)$$

Finally, $g(\theta,\lambda)$ can be either the parameter $\theta$ or any of the SC such as $\mathcal{R}(\theta,\lambda;t)$, $h(\theta,\lambda;t)$ and $MdTF$. The associated ACIs could be obtained correspondingly, and the details are omitted here for conciseness.

## 3. Pivotal inference

In this section, a pivotal quantity-based approach is proposed to estimate the survival characteristics and the difference in different test facilities. First, we derive some important results in the subsequent paragraphs and then proposed an algorithm to obtain the point as well as generalized confidence interval (GCI) estimates of aforementioned quantities.

For sample size $n_i$ and observed number of failures $m_i$, let $\mathcal{T}_i = \{\mathcal{T}_{i1}, \mathcal{T}_{i2}, \cdots, \mathcal{T}_{im_i}\}; i = 1, 2, \cdots, k$, be the progressive Type-II censored ordered sample from IER$(\theta_i,\lambda)$ distribution with censoring scheme $r_i = (r_{i1}, r_{i2}, \cdots, r_{im_i})$. Also, consider that the transformations

$$\mathcal{U}_{ij} = -\log\left(1 - \mathcal{F}(\mathcal{T}_{ij};\theta_i,\lambda)\right) = -\theta_i \log\left(1 - e^{\lambda \mathcal{T}_{ij}^2}\right)$$

for $i = 1, 2, \cdots, k$; $j = 1, 2, \cdots, m_i$. Then, it can be easily derived that the order statistics $\{\mathcal{U}_{i1}, \mathcal{U}_{i2}, \cdots, \mathcal{U}_{im_i}\}$ is a progressively Type-II censored sample from a standard exponential distribution. Further, for $i = 1, 2, \cdots k$, consider the following transformations

$$\begin{aligned}
\mathcal{V}_{i1} &= n_i \mathcal{U}_{i1} \\
\mathcal{V}_{i2} &= [n_i - (r_{i1}+1)][\mathcal{U}_{i2} - \mathcal{U}_{i1}] \\
\cdots &= \quad \cdots \\
\mathcal{V}_{im_i} &= \left[n_i - \sum_{l=1}^{m_i-1}(r_{il}+1)\right][\mathcal{U}_{im_i} - \mathcal{U}_{im_i-1}]
\end{aligned} \tag{14}$$

Then, using Jacobian transformation, it can be easily derived that the ordered random variables $\mathcal{V}_{i1}, \mathcal{V}_{i2}, \cdots, \mathcal{V}_{im_i}$ are independent and identically distributed from the standard exponential distribution. Now, for $i = 1, 2, \cdots k$, let

$$\mathcal{W}_{ij} = \sum_{l=1}^{j} \mathcal{V}_{il} = -\theta_i \left[\sum_{l=1}^{j-1}(r_{il}+1)\log\left(1 - e^{-\lambda \mathcal{T}_{il}^{-2}}\right) + \left(n_i - \sum_{l=1}^{j-1}(r_{il}+1)\right)\log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)\right]$$

and

$$\mathcal{Q}_{i1} = \frac{\mathcal{W}_{i1}}{\mathcal{W}_{im_i}}, \mathcal{Q}_{i2} = \frac{\mathcal{W}_{i2}}{\mathcal{W}_{im_i}}, \cdots, \mathcal{Q}_{im_i-1} = \frac{\mathcal{W}_{im_i-1}}{\mathcal{W}_{im_i}}$$

then one could observe that the random variables $\mathcal{Q}_{i1}, \mathcal{Q}_{i2}, \cdots, \mathcal{Q}_{im_{i-1}}$ constitutes an order statistics from a standard uniform distributon. Moreover, these random variables are independent with $\mathcal{W}_{im_i} = -\theta_i \left[\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1 - e^{-\lambda \mathcal{T}_{il}^{-2}}\right) + \left(n_i - \sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1 - e^{-\lambda \mathcal{T}_{im_i}^{-2}}\right)\right]$. Further, using the logarithmic transformation and sampling distribution theory, one can easily observe that the following quantity

$$\mathcal{P}^{\mathcal{T}_i}(\lambda) = -2\sum_{j=1}^{m_i-1}\log(\mathcal{Q}_{il})$$

$$= -2\sum_{j=1}^{m_i-1}\log\left[\frac{\sum_{l=1}^{j-1}(r_{il}+1)\log\left(1 - e^{-\lambda \mathcal{T}_{il}^{-2}}\right) + \left(n_i - \sum_{l=1}^{j-1}(r_{il}+1)\right)\log\left(1 - e^{-\lambda \mathcal{T}_{ij}^{-2}}\right)}{\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1 - e^{-\lambda \mathcal{T}_{il}^{-2}}\right) + \left(n_i - \sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1 - e^{-\lambda \mathcal{T}_{im_i}^{-2}}\right)}\right]$$

is a chi-square distribution with degree of freedom $2(m_i - 1)$, and the quantity

$$\mathcal{R}^{\mathcal{T}_i}(\lambda, \theta_i) = -2\theta_i \left[\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1 - e^{-\lambda \mathcal{T}_{il}^{-2}}\right) + \left(n_i - \sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1 - e^{-\lambda \mathcal{T}_{im_i}^{-2}}\right)\right]$$

is a chi-square distribution with degree of freedom $2m_i$.

**Theorem 3.1.** For each $i; i = 1, 2, \cdots, k$, the pivotal quantity $\mathcal{P}^{\mathcal{T}_i}(\lambda)$ increases in $\lambda$.

*Proof.* It is to be noted that we have

$$\left[\frac{\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)+\left(n_i-\sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1-e^{-\lambda\mathcal{T}_{im_i}^{-2}}\right)}{\sum_{l=1}^{j-1}(r_{il}+1)\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)+\left(n_i-\sum_{l=1}^{j-1}(r_{il}+1)\right)\log\left(1-e^{-\lambda\mathcal{T}_{ij}^{-2}}\right)}\right]$$

$$=\left[1+\frac{\sum_{l=j+1}^{m_i-1}(r_{il}+1)\frac{\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)}{\log\left(1-e^{-\lambda\mathcal{T}_{ij}^{-2}}\right)}+\left(n_i-\sum_{l=1}^{m_i-1}(r_{il}+1)\right)\frac{\log\left(1-e^{-\lambda\mathcal{T}_{im_i}^{-2}}\right)}{\log\left(1-e^{-\lambda\mathcal{T}_{ij}^{-2}}\right)}-\left(n_i-\sum_{l=1}^{j-1}(r_{il}+1)\right)}{\sum_{l=1}^{j-1}(r_{il}+1)\frac{\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)}{\log\left(1-e^{-\lambda\mathcal{T}_{ij}^{-2}}\right)}+\left(n_i-\sum_{l=1}^{j-1}(r_{il}+1)\right)}\right]$$

for $i=1,2,\cdots,k$ and $j=1,2,\cdots,m_i$. Further note that, for $0<a<b<\infty$, $g(t)=(\log(1-b^t)/\log(1-a^t))$ increases in $t$ for $t>0$ and $\lim_{t\to 0}g(t)=1$ and $\lim_{t\to\infty}g(t)=\infty$. From these results, we find that denominators in the first results decreases as $\lambda$ increase while respective numerators of above both quantities increases with $\lambda$. So, it can be easily concluded that the pivotal quantity $\mathcal{P}^{\mathcal{T}_i}(\lambda)$ increases in $\lambda$ and $\lim_{\lambda\to 0}\mathcal{P}^{\mathcal{T}_i}(\lambda)=0$ and $\lim_{\lambda\to\infty}\mathcal{P}^{\mathcal{T}_i}(\lambda)=\infty$.□

Now, consider the pivotal quantity

$$\mathcal{P}(\lambda)=\sum_{i=1}^{k}\mathcal{P}^{\mathcal{T}_i}(\lambda)$$

$$=2\sum_{i=1}^{k}\sum_{j=1}^{m_i-1}\log\left[\frac{\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)+\left(n_i-\sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1-e^{-\lambda\mathcal{T}_{im_i}^{-2}}\right)}{\sum_{l=1}^{j-1}(r_{il}+1)\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)+\left(n_i-\sum_{l=1}^{j-1}(r_{il}+1)\right)\log\left(1-e^{-\lambda\mathcal{T}_{ij}^{-2}}\right)}\right]$$

Since $\mathcal{P}^{\mathcal{T}_i}(\lambda)$ are independent from one another for $i=1,2,\cdots,k$, therefore the pivotal quantity $\mathcal{P}(\lambda)$ is chi-squared distribution with degree of freedom $2\sum_{i=1}^{k}(m_i-1)$. Also, using Theorem 3.1, it can be shown that $\mathcal{P}(\lambda)$ increases with increase in $\lambda$.

Note that the equation $\mathcal{P}(\lambda)=\lambda^*$ has a unique solution for $\lambda^*\sim\chi^2_{2\sum_{i=1}^{k}(m_i-1)}$, denoted as $g(\lambda^*,T)$. Moreover $\theta_i$ can be written as

$$\theta_i=\frac{\mathcal{R}^{\mathcal{T}_i}}{\mathcal{G}^{\mathcal{T}_i}(\lambda)},$$

where $\mathcal{G}^{\mathcal{T}_i}(\lambda)=-2\left[\sum_{l=1}^{m_i-1}(r_{il}+1)\log\left(1-e^{-\lambda\mathcal{T}_{il}^{-2}}\right)+\left(n_i-\sum_{l=1}^{m_i-1}(r_{il}+1)\right)\log\left(1-e^{-\lambda\mathcal{T}_{im_i}^{-2}}\right)\right]$.

Then, a generalized pivotal quantity for $\theta_i, i=1,2,\cdots,k$, can be obtained by $\frac{\mathcal{R}^{\mathcal{T}_i}}{\mathcal{G}^{\mathcal{T}_i}(g(\lambda^*,T))}$ (see (Weerahandi, 2004)). Based on these generalized pivotal quantities, an algorithm is proposed next to obtain the estimates of the SC and the DDTF.

### 3.1. Algorithm to find generalized estimates for SC and DDTF using pivotal method

Step 1: Set $s=1$.

Step 2: Generate a random number $\lambda^*$ from $\chi^2_{2\sum_{i=1^k}(m_i-1)}$ distribution. Then, obtain an observation $\lambda^s$ for $\lambda$ by solving the equation $\mathcal{P}(\lambda)=\lambda^*$.

Step 3: For each $i = 1, 2, \cdots, k$, generate a random number $\mathcal{R}^{\mathcal{T}_i}$ from $\chi^2_{2m_i}$ distribution. Then, obtain an observation $\theta_i^s$ for $\theta_i$ using $\theta_i^s = \frac{\mathcal{R}^{\mathcal{T}_i}}{\mathcal{G}^{\mathcal{T}_i}(\lambda^s)}$.

Step 4: Compute

$$\theta^{(s)} = \frac{\sum_{i=1}^{m} \frac{1}{\mathrm{Var}(\theta_i|\mathcal{T})} \theta_i^{(s)}}{\sum_{i=1}^{m} \frac{1}{\mathrm{Var}(\theta_i|\mathcal{T})}}$$

here $\mathrm{Var}(\theta_i|\mathcal{T}) = \frac{1}{s}\sum_{l=1}^{s}\left(\theta_i^{(l)} - \frac{1}{s}\sum_{l=1}^{s}\theta_i^{(l)}\right)^2$. Obtain

$$\mathcal{R}(t;\lambda^{(s)},\theta^{(s)}) = \left(1 - e^{-\lambda^{(s)}t^{-2}}\right)^{\theta^{(s)}}, h(t;\theta^{(s)},\lambda^{(s)}) = 2\theta^{(s)}\lambda^{(s)}t^{-3}e^{-\lambda^{(s)}t^{-2}}\left(1 - e^{-\lambda^{(s)}t^{-2}}\right)^{-1}$$

and

$$\mathrm{MdTF} = \sqrt{\frac{-\lambda^{(s)}}{\log(1 - (0.5)^{1/\theta^{(s)}})}} \tag{15}$$

Step 5: Set $s = s + 1$.

Step 6: Repeat steps 2–5, $N$ times to obtain pivotal samples as $\kappa^{(1)}, \kappa^{(2)}, \ldots, \kappa^{(N)}$ where $\kappa$ could be any of $\theta_i$, $\theta$, $\lambda$, $\mathcal{R}$, $h$ and MdTF. Then, we obtained generalized estimates of $\kappa$ as

$$\hat{\kappa}_B = E(\kappa|\mathcal{T}) = \frac{1}{N}\sum_{i=1}^{N}\kappa^{(i)}$$

Step 7: Compute the variances of $\kappa$ as

$$\mathrm{Var}(\kappa|\mathcal{T}) = \frac{1}{N}\sum_{i=1}^{N}\left(\kappa^{(i)} - \hat{\kappa}_B\right)^2$$

Step 8: To compute the generalized confidence intervals of $\kappa$, order $\kappa^{(1)}, \kappa^{(2)}, \ldots, \kappa^{(N)}$ as follows $\kappa_{(1)}, \kappa_{(2)}, \ldots, \kappa_{(N)}$. Then, the approximate $100(1-2\delta)\%$ generalized confidence intervals of $\kappa$ become $\left(\kappa_{(N\delta)}, \kappa_{(N(1-\delta))}\right)$.

The pivotal inferences on three SC and the DDTF are derived using $N$ samples. The differences of generalized confidence intervals of $\theta_i (i = 1, 2, \ldots, k)$ are deemed as pivotal inference on the DDTF.

## 4. Numerical results

In this section, we performed an extensive simulation study to tackle and compare the performances of proposed estimates for model parameters along with associated survival characteristics and DDTF. In order to do that, progressively censored samples are generated using the ideas of (Balakrishnan & Sandhu, 1995) and to simulate the effect of the DDTF, we add a noise to the generated samples using normal distribution, i.e. $N(0, 0.001 * i)$ for $i$-th test facilities. In this way, block progressive censored data is generated with respect to test facilities. Based on 5000 replications of such samples, the performances of the proposed estimates are compared using criteria quantities such as average bias, variances and interval lengths associated with estimates. In addition, there needed some specific values of the parameters such as $\theta$, $\lambda$, total sample size $n$, and block size (number of test facilities) $k$ to generate block progressive censored samples. Also, for $i$-th test facility, we further need to specify sample size $n_i$, effect sample size $m_i$. Thus, we specified $(\lambda, \theta) = (2, 1.5)$ and the corresponding total sample size, block size, sample sizes and effective sample sizes with respect to test facilities are specified as reported in Table 1. Further, in each case of these test facilities, three different cases of censoring schemes are considered as listed below:

$$\begin{aligned}
&\text{(I)} \quad r_1 = r_2 = \cdots = r_{m_i-1} = 0\, r_{m_i} = (n_i - m_i),\\
&\text{(II)} \quad r_1 = (n_i - m_i), r_2 = \cdots = r_{m_i-1} = 0 = r_{m_i},\\
&\text{(III)} \quad r_1 = (n_i - m_i)/2,\ r_2 = \cdots = r_{m_i-1} = 0,\ r_{m_i} = (n_i - m_i)/2.
\end{aligned}$$

Now, for each cases of the above settings, the estimates of the model parameters, reliability characteristics and the DDTF are obtained using maximum likelihood and pivotal quantity-based methods. The mission time is taken to be 1 unit with respect to the estimates of the survival characteristics. All the obtained estimated results are reported in the Tables 2–13. From the tabulated results, we observe that the biases, variances and interval lengths in the estimates obtained for $(\lambda, \theta)$, $R(x = 1)$, $h(x = 1)$, and MdTF for both the approaches decreases as the total sample size, sample sizes and effective sample sizes in each block or combination of any two or three increases. This demonstrates the consistency in the derived estimates, which indicate good estimations in the planned situations. Also, the estimates of $\theta_i$ with respect to $i$th test facility exhibit distinct biases and variances with other two test facilities indicating difference in test facilities. Moreover, these biases and variances are notably larger than those associated with $\theta$ indicating significant effect of test facilities. Thus, it makes sense to describe the $i$-th test facility impact using $\theta_i$. Further, it is clear that, in almost all of the cases, the pivotal procedure yielded estimates that has smaller average biases, variances, and interval lengths than those produced using the maximum likelihood method. As a result, the suggested pivotal-based inference is preferable than the inference made using the maximum likelihood method. Moreover, the ACIs and GCIs for $\theta_i; i = 1, 2, \cdots, k$, have some overlap, demonstrating the consistency of various test facilities, and also, they are not exactly the same, demonstrating the DDTF. Also, the ALs from the pivotal and likelihood methods are still far off from one another, clearly indicating the impact of the DDTF, which may not be insignificant from a practical standpoint. This shows that the DDTF cannot be ignored. Based on the entire discussion, we recommend leveraging pivotal-based inferences to estimate the survival characteristics and the DDTF for block progressive censored data.

## 5. Real data analysis

In this section, two real data sets are analyzed to illustrate and support the proposed study for application point of view.

### 5.1. Real data 1

This data set represents the number of fatalities in coal mining disasters in Britain, which claimed 10 or more lives [see (Hand et al., 1993; Jarrett, 1979)]. Here, it may be of interest to study the differences in number of fatalities during the different periods. So, the different periods can be

Table 1. Progressive Type-II block censoring schemes.

| n | Blocks k | Sample size $(n_1, n_2, \ldots, n_k)$ | Effective sample size $(m_1, m_2, \ldots, m_k)$ |
|---|---|---|---|
| 60 | 3 | $N_1$ = 20, 20, 20 | $M_1$ = 15, 18, 17<br>$M_2$ = 12, 14, 18 |
| 80 | 3 | $N_2$ = 25, 30, 25 | $M_3$ = 20, 25, 18<br>$M_4$ = 18 24, 20 |
| 100 | 3 | $N_3$ = 30, 45, 25 | $M_5$ = 25, 38, 19<br>$M_6$ = 22, 35, 16 |
| 60 | 4 | $N_4$ = 15, 15, 15, 15 | $M_7$ =12, 13, 14, 11<br>$M_8$ = 10, 12, 8, 9 |
| 80 | 4 | $N_5$ = 20, 20, 20, 20 | $M_9$ =15, 18, 14, 12<br>$M_{10}$ = 18, 15, 16, 10 |
| 100 | 4 | $N_6$ = 25, 30, 25, 20 | $M_{11}$ =20, 25, 20, 15<br>$M_{12}$ = 18, 22, 20, 14 |

**Table 2.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_1$ and $M_1$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 60 | I | $\lambda$ | 0.3129 | 0.1267 | 1.4782 | 2.8599 | 1.3817 | 0.2795 | 0.1232 | 1.4082 | 2.7677 | 1.3594 |
| | | $\theta_1$ | 0.5078 | 0.4429 | 0.6450 | 2.9814 | 2.2315 | 0.4607 | 0.1764 | 1.1288 | 2.5097 | 1.3809 |
| | | $\theta_2$ | 0.4457 | 0.3005 | 0.7499 | 2.7461 | 1.9962 | 0.4045 | 0.1054 | 1.1619 | 2.7461 | 1.5842 |
| | | $\theta_3$ | 0.4604 | 0.3403 | 0.7236 | 2.8366 | 2.1130 | 0.4170 | 0.1244 | 1.1587 | 2.3722 | 1.2135 |
| | | $\theta$ | 0.2939 | 0.1417 | 0.8929 | 2.3043 | 1.4115 | 0.2765 | 0.0826 | 1.0694 | 2.1295 | 1.0601 |
| | | $R(x)$ | 0.0366 | 0.1276 | 0.1491 | 1.0000 | 0.8472 | 0.0331 | 0.0011 | 0.1491 | 0.8693 | 0.7202 |
| | | $h(x)$ | 0.1255 | 0.0248 | 0.5740 | 1.1856 | 0.6116 | 0.1147 | 0.0030 | 0.7741 | 0.9666 | 0.1925 |
| | | MdTF | 0.0754 | 0.0532 | 1.0065 | 1.9062 | 0.8997 | 0.0703 | 0.0007 | 1.3897 | 1.4893 | 0.0995 |
| | II | $\lambda$ | 0.3026 | 0.1190 | 1.4940 | 2.8330 | 1.3390 | 0.2731 | 0.1166 | 1.4402 | 2.7616 | 1.3213 |
| | | $\theta_1$ | 0.4406 | 0.3190 | 0.7071 | 2.7738 | 1.9961 | 0.4091 | 0.0911 | 1.1941 | 2.2405 | 1.0464 |
| | | $\theta_2$ | 0.4012 | 0.2644 | 0.7777 | 2.6784 | 1.9006 | 0.3687 | 0.0801 | 1.2054 | 2.6784 | 1.4730 |
| | | $\theta_3$ | 0.4366 | 0.2844 | 0.7563 | 2.7081 | 1.9518 | 0.4054 | 0.0852 | 1.2019 | 2.2174 | 1.0155 |
| | | $\theta$ | 0.2620 | 0.1114 | 0.9391 | 2.2033 | 1.2643 | 0.2525 | 0.0562 | 1.1187 | 2.0019 | 0.8832 |
| | | $R(x)$ | 0.0400 | 0.1006 | 0.2192 | 1.0000 | 0.8075 | 0.0364 | 0.0012 | 0.2192 | 0.8740 | 0.6549 |
| | | $h(x)$ | 0.1426 | 0.0261 | 0.5617 | 1.1888 | 0.6271 | 0.1321 | 0.0047 | 0.7413 | 0.9912 | 0.2500 |
| | | MdTF | 0.0865 | 0.0464 | 1.0418 | 1.8819 | 0.8401 | 0.0808 | 0.0013 | 1.3746 | 1.5151 | 0.1405 |
| | III | $\lambda$ | 0.3059 | 0.1220 | 1.4853 | 2.8413 | 1.3559 | 0.2766 | 0.1189 | 1.4258 | 2.7624 | 1.3366 |
| | | $\theta_1$ | 0.4908 | 0.3735 | 0.6838 | 2.8689 | 2.0966 | 0.4536 | 0.1269 | 1.1685 | 2.3640 | 1.1956 |
| | | $\theta_2$ | 0.4507 | 0.3053 | 0.7723 | 2.7590 | 1.9867 | 0.4152 | 0.1040 | 1.1999 | 2.7590 | 1.5591 |
| | | $\theta_3$ | 0.4430 | 0.3044 | 0.7471 | 2.7635 | 2.0164 | 0.4078 | 0.0983 | 1.1892 | 2.2833 | 1.0941 |
| | | $\theta$ | 0.2812 | 0.1261 | 0.9226 | 2.2577 | 1.3350 | 0.2702 | 0.0683 | 1.1018 | 2.0688 | 0.9671 |
| | | $R(x)$ | 0.0378 | 0.1136 | 0.1849 | 1.0000 | 0.8374 | 0.0348 | 0.0012 | 0.1849 | 0.8715 | 0.6865 |
| | | $h(x)$ | 0.1348 | 0.0256 | 0.5703 | 1.1917 | 0.6214 | 0.1247 | 0.0039 | 0.7590 | 0.9830 | 0.2240 |
| | | MdTF | 0.0801 | 0.0491 | 1.0243 | 1.8887 | 0.8644 | 0.0749 | 0.0010 | 1.3786 | 1.5009 | 0.1223 |

**Table 3.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_1$ and $M_2$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 60 | I | $\lambda$ | 0.3412 | 0.1446 | 1.4532 | 2.9265 | 1.4733 | 0.2993 | 0.1396 | 1.3740 | 2.8160 | 1.4420 |
| | | $\theta_1$ | 0.6818 | 0.9017 | 0.4592 | 3.4616 | 2.8857 | 0.6111 | 0.4570 | 1.0721 | 2.9797 | 1.9076 |
| | | $\theta_2$ | 0.5359 | 0.5087 | 0.5759 | 3.0841 | 2.5082 | 0.4775 | 0.2145 | 1.0768 | 3.0841 | 2.0073 |
| | | $\theta_3$ | 0.4487 | 0.3211 | 0.7376 | 2.7978 | 2.0603 | 0.4024 | 0.1194 | 1.1407 | 2.3394 | 1.1987 |
| | | $\theta$ | 0.3358 | 0.1911 | 0.8169 | 2.4286 | 1.6117 | 0.3178 | 0.1226 | 1.0107 | 2.2711 | 1.2604 |
| | | $R(x)$ | 0.0389 | 0.1692 | 0.0598 | 1.0000 | 0.9285 | 0.0351 | 0.0010 | 0.0598 | 0.8679 | 0.8082 |
| | | $h(x)$ | 0.1418 | 0.0277 | 0.5554 | 1.1988 | 0.6435 | 0.1321 | 0.0021 | 0.7873 | 0.9417 | 0.1544 |
| | | MdTF | 0.0852 | 0.0644 | 0.9661 | 1.9549 | 0.9888 | 0.0812 | 0.0003 | 1.4168 | 1.4802 | 0.0635 |
| | II | $\lambda$ | 0.3042 | 0.1252 | 1.4535 | 2.8258 | 1.3723 | 0.2797 | 0.1227 | 1.3977 | 2.7548 | 1.3571 |
| | | $\theta_1$ | 0.5262 | 0.4642 | 0.6080 | 3.0086 | 2.3334 | 0.4842 | 0.1374 | 1.1812 | 2.3900 | 1.2088 |
| | | $\theta_2$ | 0.4682 | 0.3653 | 0.6752 | 2.8584 | 2.1832 | 0.4316 | 0.1092 | 1.1823 | 2.8584 | 1.6761 |
| | | $\theta_3$ | 0.4110 | 0.2624 | 0.7602 | 2.6492 | 1.8890 | 0.3839 | 0.0823 | 1.1753 | 2.1905 | 1.0152 |
| | | $\theta$ | 0.2795 | 0.1270 | 0.8974 | 2.2406 | 1.3432 | 0.2688 | 0.0643 | 1.0881 | 2.0287 | 0.9406 |
| | | $R(x)$ | 0.0402 | 0.1129 | 0.1825 | 1.0000 | 0.8473 | 0.0375 | 0.0013 | 0.1825 | 0.8718 | 0.6894 |
| | | $h(x)$ | 0.1463 | 0.0301 | 0.5505 | 1.2218 | 0.6713 | 0.1370 | 0.0044 | 0.7524 | 0.9908 | 0.2384 |
| | | MdTF | 0.0879 | 0.0509 | 1.0171 | 1.8965 | 0.8793 | 0.0831 | 0.0012 | 1.3740 | 1.5097 | 0.1350 |
| | III | $\lambda$ | 0.3123 | 0.1343 | 1.4445 | 2.8667 | 1.4221 | 0.2809 | 0.1303 | 1.3749 | 2.7727 | 1.3977 |
| | | $\theta_1$ | 0.5575 | 0.5402 | 0.5289 | 3.1049 | 2.4768 | 0.5103 | 0.2058 | 1.0890 | 2.5722 | 1.4832 |
| | | $\theta_2$ | 0.5040 | 0.4279 | 0.6281 | 2.9836 | 2.3555 | 0.4544 | 0.1539 | 1.1271 | 2.9836 | 1.8565 |
| | | $\theta_3$ | 0.4408 | 0.2978 | 0.7486 | 2.7327 | 1.9841 | 0.4024 | 0.1037 | 1.1551 | 2.2723 | 1.1172 |
| | | $\theta$ | 0.3143 | 0.1573 | 0.8532 | 2.3332 | 1.4799 | 0.3006 | 0.0914 | 1.0404 | 2.1487 | 1.1083 |
| | | $R(x)$ | 0.0375 | 0.1393 | 0.1192 | 1.0000 | 0.9040 | 0.0343 | 0.0011 | 0.1192 | 0.8692 | 0.7499 |
| | | $h(x)$ | 0.1393 | 0.0285 | 0.5570 | 1.2107 | 0.6538 | 0.1300 | 0.0029 | 0.7760 | 0.9626 | 0.1866 |
| | | MdTF | 0.0841 | 0.0581 | 0.9876 | 1.9274 | 0.9398 | 0.0796 | 0.0006 | 1.3957 | 1.4911 | 0.0954 |

Table 4. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_2$ and $M_3$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 80 | I | $\lambda$ | 0.2670 | 0.0968 | 1.5209 | 2.7311 | 1.2102 | 0.2448 | 0.0946 | 1.4664 | 2.6608 | 1.1944 |
| | | $\theta_1$ | 0.3929 | 0.2511 | 0.7745 | 2.6207 | 1.7779 | 0.3659 | 0.0892 | 1.1616 | 2.2192 | 1.0575 |
| | | $\theta_2$ | 0.3614 | 0.2029 | 0.8429 | 2.5174 | 1.6746 | 0.3351 | 0.0775 | 1.1663 | 2.5174 | 1.3512 |
| | | $\theta_3$ | 0.4317 | 0.3060 | 0.7230 | 2.7284 | 2.0055 | 0.4022 | 0.1155 | 1.1444 | 2.3180 | 1.1736 |
| | | $\theta$ | 0.2636 | 0.1108 | 0.9423 | 2.1976 | 1.2553 | 0.2547 | 0.0669 | 1.0941 | 2.0556 | 0.9615 |
| | | $R(x)$ | 0.0317 | 0.0996 | 0.2206 | 1.0000 | 0.7968 | 0.0292 | 0.0008 | 0.2206 | 0.8610 | 0.6404 |
| | | $h(x)$ | 0.1150 | 0.0192 | 0.6215 | 1.1600 | 0.5384 | 0.1083 | 0.0018 | 0.8065 | 0.9549 | 0.1484 |
| | | MdTF | 0.0692 | 0.0433 | 1.0441 | 1.8569 | 0.8128 | 0.0657 | 0.0003 | 1.4029 | 1.4754 | 0.0725 |
| | II | $\lambda$ | 0.2451 | 0.0882 | 1.5237 | 2.6800 | 1.1563 | 0.2294 | 0.0870 | 1.4851 | 2.6303 | 1.1452 |
| | | $\theta_1$ | 0.3588 | 0.2087 | 0.8142 | 2.5169 | 1.6345 | 0.3399 | 0.0572 | 1.2207 | 2.0764 | 0.8557 |
| | | $\theta_2$ | 0.3271 | 0.1714 | 0.8824 | 2.4358 | 1.5534 | 0.3103 | 0.0527 | 1.2258 | 2.4358 | 1.2100 |
| | | $\theta_3$ | 0.4030 | 0.2444 | 0.7863 | 2.5996 | 1.8132 | 0.3832 | 0.0663 | 1.2291 | 2.1229 | 0.8938 |
| | | $\theta$ | 0.2300 | 0.0832 | 0.9993 | 2.0989 | 1.0996 | 0.2255 | 0.0418 | 1.1572 | 1.9261 | 0.7689 |
| | | $R(x)$ | 0.0318 | 0.0750 | 0.2918 | 1.0000 | 0.7486 | 0.0301 | 0.0010 | 0.2918 | 0.8642 | 0.5725 |
| | | $h(x)$ | 0.1138 | 0.0208 | 0.6175 | 1.1791 | 0.5616 | 0.1079 | 0.0035 | 0.7790 | 0.9969 | 0.2179 |
| | | MdTF | 0.0677 | 0.0356 | 1.0775 | 1.8148 | 0.7374 | 0.0646 | 0.0010 | 1.3730 | 1.4954 | 0.1223 |
| | III | $\lambda$ | 0.2475 | 0.0908 | 1.5066 | 2.6791 | 1.1725 | 0.2322 | 0.0889 | 1.4603 | 2.6175 | 1.1572 |
| | | $\theta_1$ | 0.3766 | 0.2314 | 0.7967 | 2.5604 | 1.7113 | 0.3562 | 0.0735 | 1.1927 | 2.1359 | 0.9431 |
| | | $\theta_2$ | 0.3188 | 0.1739 | 0.8491 | 2.4095 | 1.5604 | 0.3004 | 0.0581 | 1.1751 | 2.4095 | 1.2344 |
| | | $\theta_3$ | 0.3944 | 0.2568 | 0.7512 | 2.6211 | 1.8700 | 0.3683 | 0.0806 | 1.1753 | 2.1781 | 1.0028 |
| | | $\theta$ | 0.2307 | 0.0903 | 0.9639 | 2.1092 | 1.1454 | 0.2260 | 0.0493 | 1.1155 | 1.9513 | 0.8357 |
| | | $R(x)$ | 0.0313 | 0.0817 | 0.2686 | 1.0000 | 0.7941 | 0.0295 | 0.0009 | 0.2686 | 0.8608 | 0.5922 |
| | | $h(x)$ | 0.1098 | 0.0201 | 0.6203 | 1.1714 | 0.5511 | 0.1028 | 0.0026 | 0.7953 | 0.9773 | 0.1820 |
| | | MdTF | 0.0660 | 0.0390 | 1.0613 | 1.8333 | 0.7720 | 0.0621 | 0.0007 | 1.3841 | 1.4848 | 0.1007 |

Table 5. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_2$ and $M_4$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 80 | I | $\lambda$ | 0.2759 | 0.0986 | 1.5244 | 2.7450 | 1.2206 | 0.2505 | 0.0965 | 1.4670 | 2.6714 | 1.2044 |
| | | $\theta_1$ | 0.4587 | 0.3270 | 0.7326 | 2.7979 | 1.9774 | 0.4235 | 0.1269 | 1.1594 | 2.3809 | 1.2216 |
| | | $\theta_2$ | 0.3674 | 0.2177 | 0.8205 | 2.5429 | 1.7224 | 0.3384 | 0.0854 | 1.1507 | 2.5429 | 1.3923 |
| | | $\theta_3$ | 0.4072 | 0.2698 | 0.7853 | 2.6840 | 1.8988 | 0.3770 | 0.0995 | 1.1774 | 2.2763 | 1.0989 |
| | | $\theta$ | 0.2652 | 0.1169 | 0.9417 | 2.2319 | 1.2902 | 0.2531 | 0.0713 | 1.0970 | 2.0894 | 0.9925 |
| | | $R(x)$ | 0.0321 | 0.1046 | 0.2048 | 1.0000 | 0.8272 | 0.0302 | 0.0008 | 0.2048 | 0.8600 | 0.6552 |
| | | $h(x)$ | 0.1151 | 0.0196 | 0.6231 | 1.1677 | 0.5447 | 0.1079 | 0.0017 | 0.8140 | 0.9582 | 0.1441 |
| | | MdTF | 0.0672 | 0.0439 | 1.0378 | 1.8557 | 0.8180 | 0.0634 | 0.0003 | 1.4006 | 1.4697 | 0.0691 |
| | II | $\lambda$ | 0.2511 | 0.0900 | 1.5296 | 2.6970 | 1.1674 | 0.2341 | 0.0886 | 1.4899 | 2.6439 | 1.1540 |
| | | $\theta_1$ | 0.4094 | 0.2491 | 0.7899 | 2.6237 | 1.7629 | 0.3867 | 0.0684 | 1.2339 | 2.1442 | 0.9103 |
| | | $\theta_2$ | 0.3301 | 0.1742 | 0.8608 | 2.4246 | 1.5638 | 0.3128 | 0.0526 | 1.2095 | 2.4246 | 1.2151 |
| | | $\theta_3$ | 0.3566 | 0.2100 | 0.8184 | 2.5288 | 1.7104 | 0.3349 | 0.0574 | 1.2258 | 2.0841 | 0.8583 |
| | | $\theta$ | 0.2272 | 0.0831 | 0.9914 | 2.0926 | 1.1013 | 0.2208 | 0.0413 | 1.1506 | 1.9178 | 0.7671 |
| | | $R(x)$ | 0.0326 | 0.0752 | 0.2924 | 1.0000 | 0.7519 | 0.0303 | 0.0010 | 0.2924 | 0.8660 | 0.5736 |
| | | $h(x)$ | 0.1166 | 0.0208 | 0.6074 | 1.1693 | 0.5619 | 0.1100 | 0.0036 | 0.7698 | 0.9893 | 0.2195 |
| | | MdTF | 0.0699 | 0.0364 | 1.0794 | 1.8242 | 0.7447 | 0.0659 | 0.0010 | 1.3777 | 1.5003 | 0.1226 |
| | III | $\lambda$ | 0.2701 | 0.0948 | 1.5292 | 2.7262 | 1.1970 | 0.2495 | 0.0929 | 1.4798 | 2.6633 | 1.1835 |
| | | $\theta_1$ | 0.4148 | 0.2763 | 0.7586 | 2.6691 | 1.8255 | 0.3867 | 0.0904 | 1.1870 | 2.2231 | 1.0360 |
| | | $\theta_2$ | 0.3573 | 0.1960 | 0.8436 | 2.4933 | 1.6497 | 0.3343 | 0.0685 | 1.1814 | 2.4933 | 1.3118 |
| | | $\theta_3$ | 0.3821 | 0.2317 | 0.8003 | 2.5789 | 1.7785 | 0.3586 | 0.0731 | 1.1959 | 2.1533 | 0.9574 |
| | | $\theta$ | 0.2521 | 0.0976 | 0.9668 | 2.1517 | 1.1849 | 0.2440 | 0.0541 | 1.1217 | 1.9929 | 0.8713 |
| | | $R(x)$ | 0.0339 | 0.0880 | 0.2539 | 1.0000 | 0.7310 | 0.0312 | 0.0009 | 0.2539 | 0.8641 | 0.6102 |
| | | $h(x)$ | 0.1217 | 0.0200 | 0.6121 | 1.1623 | 0.5502 | 0.1138 | 0.0026 | 0.7857 | 0.9701 | 0.1844 |
| | | MdTF | 0.0726 | 0.0402 | 1.0615 | 1.8443 | 0.7827 | 0.0682 | 0.0006 | 1.3914 | 1.4892 | 0.0978 |

**Table 6.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_3$ and $M_5$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 100 | I | $\lambda$ | 0.2282 | 0.0731 | 1.5760 | 2.6298 | 1.0537 | 0.2116 | 0.0719 | 1.5338 | 2.5767 | 1.0429 |
| | | $\theta_1$ | 0.3397 | 0.1770 | 0.8653 | 2.4335 | 1.4707 | 0.3223 | 0.0579 | 1.2077 | 2.0727 | 0.8649 |
| | | $\theta_2$ | 0.2925 | 0.1272 | 0.9628 | 2.3117 | 1.3490 | 0.2752 | 0.0513 | 1.2065 | 2.3117 | 1.1052 |
| | | $\theta_3$ | 0.3844 | 0.2446 | 0.7859 | 2.6141 | 1.8282 | 0.3636 | 0.0760 | 1.2122 | 2.1867 | 0.9745 |
| | | $\theta$ | 0.2124 | 0.0769 | 1.0192 | 2.0811 | 1.0618 | 0.2096 | 0.0454 | 1.1490 | 1.9565 | 0.8075 |
| | | $R(x)$ | 0.0284 | 0.0697 | 0.3091 | 1.0000 | 0.7145 | 0.0268 | 0.0006 | 0.3091 | 0.8555 | 0.5464 |
| | | $h(x)$ | 0.0991 | 0.0150 | 0.6610 | 1.1374 | 0.4764 | 0.0956 | 0.0016 | 0.8211 | 0.9630 | 0.1419 |
| | | MdTF | 0.0584 | 0.0322 | 1.0931 | 1.7947 | 0.7015 | 0.0568 | 0.0003 | 1.3983 | 1.4701 | 0.0718 |
| | II | $\lambda$ | 0.2138 | 0.0684 | 1.5714 | 2.5909 | 1.0196 | 0.2033 | 0.0674 | 1.5393 | 2.5517 | 1.0125 |
| | | $\theta_1$ | 0.3016 | 0.1472 | 0.8804 | 2.3316 | 1.3520 | 0.2897 | 0.0374 | 1.2370 | 1.9468 | 0.7099 |
| | | $\theta_2$ | 0.2547 | 0.1045 | 0.9796 | 2.2146 | 1.2350 | 0.2436 | 0.0341 | 1.2358 | 2.2146 | 0.9787 |
| | | $\theta_3$ | 0.3445 | 0.2014 | 0.8149 | 2.4949 | 1.6800 | 0.3287 | 0.0452 | 1.2587 | 2.0221 | 0.7634 |
| | | $\theta$ | 0.1941 | 0.0599 | 1.0528 | 1.9956 | 0.9427 | 0.1921 | 0.0298 | 1.1873 | 1.8470 | 0.6597 |
| | | $R(x)$ | 0.0288 | 0.0546 | 0.3636 | 1.0000 | 0.7030 | 0.0275 | 0.0008 | 0.3636 | 0.8593 | 0.4957 |
| | | $h(x)$ | 0.1022 | 0.0159 | 0.6534 | 1.1452 | 0.4918 | 0.0986 | 0.0028 | 0.7921 | 0.9919 | 0.1998 |
| | | MdTF | 0.0615 | 0.0278 | 1.1195 | 1.7708 | 0.6513 | 0.0596 | 0.0008 | 1.3796 | 1.4910 | 0.1114 |
| | III | $\lambda$ | 0.2256 | 0.0703 | 1.5630 | 2.5957 | 1.0326 | 0.2131 | 0.0691 | 1.5270 | 2.5483 | 1.0213 |
| | | $\theta_1$ | 0.3075 | 0.1593 | 0.8750 | 2.3772 | 1.4180 | 0.2927 | 0.0455 | 1.2241 | 2.0010 | 0.7769 |
| | | $\theta_2$ | 0.2652 | 0.1107 | 0.9593 | 2.2216 | 1.2623 | 0.2556 | 0.0397 | 1.2061 | 2.2216 | 1.0155 |
| | | $\theta_3$ | 0.3795 | 0.2242 | 0.7955 | 2.5408 | 1.7452 | 0.3611 | 0.0615 | 1.2271 | 2.0947 | 0.8676 |
| | | $\theta$ | 0.2081 | 0.0671 | 1.0332 | 2.0251 | 0.9918 | 0.2082 | 0.0369 | 1.1641 | 1.8898 | 0.7257 |
| | | $R(x)$ | 0.0293 | 0.0611 | 0.3396 | 1.0000 | 0.7494 | 0.0280 | 0.0007 | 0.3396 | 0.8559 | 0.5164 |
| | | $h(x)$ | 0.1032 | 0.0155 | 0.6590 | 1.1433 | 0.4844 | 0.1004 | 0.0021 | 0.8100 | 0.9789 | 0.1690 |
| | | MdTF | 0.0618 | 0.0300 | 1.1056 | 1.7821 | 0.6765 | 0.0606 | 0.0006 | 1.3873 | 1.4799 | 0.0925 |

**Table 7.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_3$ and $M_6$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 100 | I | $\lambda$ | 0.2511 | 0.0816 | 1.5582 | 2.6701 | 1.1119 | 0.2326 | 0.0798 | 1.5071 | 2.6050 | 1.0979 |
| | | $\theta_1$ | 0.3867 | 0.2370 | 0.8128 | 2.6069 | 1.6795 | 0.3599 | 0.0875 | 1.1812 | 2.2306 | 1.0494 |
| | | $\theta_2$ | 0.3300 | 0.1576 | 0.9274 | 2.4141 | 1.4866 | 0.3082 | 0.0696 | 1.1782 | 2.4141 | 1.2358 |
| | | $\theta_3$ | 0.4472 | 0.3348 | 0.6861 | 2.7849 | 2.0988 | 0.4168 | 0.1182 | 1.1538 | 2.3390 | 1.1852 |
| | | $\theta$ | 0.2577 | 0.1023 | 0.9725 | 2.1845 | 1.2121 | 0.2494 | 0.0650 | 1.1084 | 2.0620 | 0.9536 |
| | | $R(x)$ | 0.0294 | 0.0911 | 0.2416 | 1.0000 | 0.7492 | 0.0277 | 0.0006 | 0.2416 | 0.8533 | 0.6118 |
| | | $h(x)$ | 0.1055 | 0.0167 | 0.6529 | 1.1547 | 0.5018 | 0.1018 | 0.0010 | 0.8403 | 0.9472 | 0.1069 |
| | | MdTF | 0.0626 | 0.0382 | 1.0602 | 1.8241 | 0.7638 | 0.0615 | 0.0001 | 1.4135 | 1.4580 | 0.0446 |
| | II | $\lambda$ | 0.2173 | 0.0735 | 1.5565 | 2.6131 | 1.0566 | 0.2035 | 0.0727 | 1.5242 | 2.5721 | 1.0480 |
| | | $\theta_1$ | 0.3540 | 0.1846 | 0.8568 | 2.4582 | 1.4969 | 0.3381 | 0.0486 | 1.2493 | 2.0370 | 0.7877 |
| | | $\theta_2$ | 0.2733 | 0.1187 | 0.9613 | 2.2657 | 1.3044 | 0.2620 | 0.0397 | 1.2302 | 2.2657 | 1.0355 |
| | | $\theta_3$ | 0.3784 | 0.2414 | 0.7423 | 2.5608 | 1.8186 | 0.3650 | 0.0534 | 1.2315 | 2.0418 | 0.8103 |
| | | $\theta$ | 0.1972 | 0.0676 | 1.0238 | 2.0210 | 0.9972 | 0.1986 | 0.0335 | 1.1663 | 1.8604 | 0.6941 |
| | | $R(x)$ | 0.0302 | 0.0610 | 0.3401 | 1.0000 | 0.7011 | 0.0288 | 0.0008 | 0.3401 | 0.8609 | 0.5207 |
| | | $h(x)$ | 0.1104 | 0.0177 | 0.6368 | 1.1553 | 0.5185 | 0.1073 | 0.0028 | 0.7864 | 0.9835 | 0.1971 |
| | | MdTF | 0.0675 | 0.0305 | 1.1070 | 1.7897 | 0.6827 | 0.0661 | 0.0008 | 1.3851 | 1.4952 | 0.1101 |
| | III | $\lambda$ | 0.2368 | 0.0775 | 1.5529 | 2.6368 | 1.0839 | 0.2226 | 0.0765 | 1.5100 | 2.5845 | 1.0745 |
| | | $\theta_1$ | 0.3624 | 0.2045 | 0.8310 | 2.5171 | 1.5765 | 0.3412 | 0.0651 | 1.2062 | 2.1244 | 0.9182 |
| | | $\theta_2$ | 0.2923 | 0.1349 | 0.9406 | 2.3266 | 1.3860 | 0.2763 | 0.0529 | 1.1964 | 2.3266 | 1.1302 |
| | | $\theta_3$ | 0.4293 | 0.2972 | 0.7189 | 2.6812 | 1.9623 | 0.4066 | 0.0895 | 1.1944 | 2.2009 | 1.0065 |
| | | $\theta$ | 0.2288 | 0.0830 | 0.9960 | 2.0955 | 1.0995 | 0.2293 | 0.0480 | 1.1329 | 1.9580 | 0.8250 |
| | | $R(x)$ | 0.0296 | 0.0747 | 0.2920 | 1.0000 | 0.7469 | 0.0284 | 0.0007 | 0.2920 | 0.8565 | 0.5645 |
| | | $h(x)$ | 0.1075 | 0.0170 | 0.6462 | 1.1539 | 0.5077 | 0.1053 | 0.0017 | 0.8155 | 0.9631 | 0.1476 |
| | | MdTF | 0.0639 | 0.0345 | 1.0819 | 1.8075 | 0.7256 | 0.0634 | 0.0004 | 1.3987 | 1.4749 | 0.0763 |

Table 8. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_4$ and $M_7$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 60 | I | $\lambda$ | 0.3242 | 0.1295 | 1.4930 | 2.8881 | 1.3951 | 0.2860 | 0.1244 | 1.3927 | 2.7551 | 1.3624 |
| | | $\theta_1$ | 0.5566 | 0.5237 | 0.5787 | 3.1085 | 2.4824 | 0.4911 | 0.1712 | 1.1403 | 2.4790 | 1.3387 |
| | | $\theta_2$ | 0.5029 | 0.4478 | 0.6261 | 2.9609 | 2.3348 | 0.4468 | 0.1430 | 1.1465 | 2.9609 | 1.8144 |
| | | $\theta_3$ | 0.4926 | 0.3855 | 0.6787 | 2.9089 | 2.2302 | 0.4355 | 0.1130 | 1.1748 | 2.3074 | 1.1327 |
| | | $\theta_4$ | 0.6347 | 0.6514 | 0.5191 | 3.2750 | 2.7560 | 0.5567 | 0.2241 | 1.1307 | 2.6186 | 1.4879 |
| | | $\theta$ | 0.2885 | 0.1353 | 0.8692 | 2.2477 | 1.3784 | 0.2781 | 0.0764 | 1.0297 | 2.0466 | 1.0169 |
| | | $R(x)$ | 0.0406 | 0.1231 | 0.1682 | 1.0000 | 0.9220 | 0.0348 | 0.0011 | 0.1682 | 0.8717 | 0.7035 |
| | | $h(x)$ | 0.1470 | 0.0239 | 0.5480 | 1.1479 | 0.5998 | 0.1305 | 0.0029 | 0.7521 | 0.9411 | 0.1890 |
| | | MdTF | 0.0907 | 0.0555 | 1.0188 | 1.9374 | 0.9187 | 0.0815 | 0.0007 | 1.4077 | 1.5079 | 0.1003 |
| | II | $\lambda$ | 0.3049 | 0.1181 | 1.4979 | 2.8321 | 1.3342 | 0.2678 | 0.1148 | 1.4149 | 2.7270 | 1.3121 |
| | | $\theta_1$ | 0.5474 | 0.4528 | 0.6387 | 3.0303 | 2.3720 | 0.4913 | 0.1173 | 1.2233 | 2.3516 | 1.1283 |
| | | $\theta_2$ | 0.4616 | 0.3650 | 0.6583 | 2.8389 | 2.1806 | 0.4190 | 0.0919 | 1.1921 | 2.8389 | 1.6468 |
| | | $\theta_3$ | 0.4848 | 0.3703 | 0.6967 | 2.8609 | 2.1642 | 0.4359 | 0.1022 | 1.2097 | 2.2594 | 1.0498 |
| | | $\theta_4$ | 0.5590 | 0.5008 | 0.5970 | 3.1047 | 2.5077 | 0.5002 | 0.1263 | 1.2201 | 2.3839 | 1.1637 |
| | | $\theta$ | 0.2559 | 0.1093 | 0.9198 | 2.1717 | 1.2519 | 0.2468 | 0.0540 | 1.0856 | 1.9525 | 0.8669 |
| | | $R(x)$ | 0.0400 | 0.0991 | 0.2270 | 1.0000 | 0.8977 | 0.0352 | 0.0012 | 0.2270 | 0.8732 | 0.6462 |
| | | $h(x)$ | 0.1422 | 0.0256 | 0.5487 | 1.1704 | 0.6218 | 0.1270 | 0.0044 | 0.7385 | 0.9779 | 0.2394 |
| | | MdTF | 0.0876 | 0.0474 | 1.0467 | 1.8959 | 0.8492 | 0.0789 | 0.0013 | 1.3811 | 1.5192 | 0.1381 |
| | III | $\lambda$ | 0.3088 | 0.1223 | 1.4901 | 2.8475 | 1.3573 | 0.2711 | 0.1184 | 1.4004 | 2.7313 | 1.3309 |
| | | $\theta_1$ | 0.5111 | 0.4292 | 0.6075 | 2.9678 | 2.3255 | 0.4565 | 0.1161 | 1.1671 | 2.3189 | 1.1518 |
| | | $\theta_2$ | 0.4896 | 0.3872 | 0.6423 | 2.8903 | 2.2480 | 0.4401 | 0.1069 | 1.1679 | 2.8903 | 1.7224 |
| | | $\theta_3$ | 0.4668 | 0.3515 | 0.6926 | 2.8429 | 2.1503 | 0.4184 | 0.0934 | 1.1956 | 2.2377 | 1.0421 |
| | | $\theta_4$ | 0.5753 | 0.5554 | 0.5581 | 3.1658 | 2.6077 | 0.5107 | 0.1673 | 1.1709 | 2.4882 | 1.3173 |
| | | $\theta$ | 0.2687 | 0.1186 | 0.8985 | 2.1982 | 1.2997 | 0.2625 | 0.0625 | 1.0594 | 1.9873 | 0.9279 |
| | | $R(x)$ | 0.0388 | 0.1076 | 0.2043 | 1.0000 | 0.8448 | 0.0343 | 0.0012 | 0.2043 | 0.8721 | 0.6678 |
| | | $h(x)$ | 0.1412 | 0.0248 | 0.5512 | 1.1630 | 0.6118 | 0.1275 | 0.0036 | 0.7472 | 0.9631 | 0.2158 |
| | | MdTF | 0.0875 | 0.0507 | 1.0341 | 1.9124 | 0.8784 | 0.0801 | 0.0010 | 1.3914 | 1.5132 | 0.1219 |

Table 9. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_4$ and $M_8$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 60 | I | $\lambda$ | 0.3914 | 0.1706 | 1.4678 | 3.0656 | 1.5977 | 0.3131 | 0.1608 | 1.3260 | 2.8718 | 1.5459 |
| | | $\theta_1$ | 0.7385 | 1.0433 | 0.3863 | 3.6374 | 3.0889 | 0.6154 | 0.4222 | 1.0469 | 2.9174 | 1.8705 |
| | | $\theta_2$ | 0.6493 | 0.6918 | 0.5485 | 3.3616 | 2.8131 | 0.5364 | 0.2577 | 1.1014 | 3.3616 | 2.2602 |
| | | $\theta_3$ | 0.9497 | 2.1590 | 0.1411 | 4.2730 | 4.1319 | 0.7813 | 1.0690 | 1.0221 | 3.4978 | 2.4757 |
| | | $\theta_4$ | 0.8565 | 1.4857 | 0.2825 | 4.0387 | 3.7562 | 0.6959 | 0.6625 | 1.0574 | 3.2790 | 2.2216 |
| | | $\theta$ | 0.3939 | 0.2698 | 0.7176 | 2.5955 | 1.8779 | 0.3561 | 0.1680 | 0.9227 | 2.3687 | 1.4460 |
| | | $R(x)$ | 0.0447 | 0.2367 | 0.0536 | 1.0000 | 0.9753 | 0.0365 | 0.0009 | 0.0536 | 0.8708 | 0.8244 |
| | | $h(x)$ | 0.1674 | 0.0299 | 0.5115 | 1.1766 | 0.6651 | 0.1495 | 0.0017 | 0.7658 | 0.9038 | 0.1380 |
| | | MdTF | 0.1017 | 0.0801 | 0.9313 | 2.0328 | 1.1015 | 0.0951 | 0.0001 | 1.4594 | 1.4940 | 0.0346 |
| | II | $\lambda$ | 0.3356 | 0.1370 | 1.4712 | 2.9056 | 1.4344 | 0.2939 | 0.1323 | 1.3802 | 2.7851 | 1.4049 |
| | | $\theta_1$ | 0.5889 | 0.5824 | 0.5275 | 3.2001 | 2.5811 | 0.5188 | 0.1558 | 1.1754 | 2.4436 | 1.2682 |
| | | $\theta_2$ | 0.5271 | 0.4709 | 0.6189 | 3.0270 | 2.4081 | 0.4681 | 0.1332 | 1.1822 | 3.0270 | 1.8448 |
| | | $\theta_3$ | 0.6797 | 0.8497 | 0.3934 | 3.4951 | 3.1017 | 0.5991 | 0.2309 | 1.1763 | 2.6124 | 1.4362 |
| | | $\theta_4$ | 0.6044 | 0.6750 | 0.4642 | 3.2690 | 2.8047 | 0.5372 | 0.1804 | 1.1634 | 2.4657 | 1.3024 |
| | | $\theta$ | 0.3027 | 0.1445 | 0.8409 | 2.2669 | 1.4260 | 0.2892 | 0.0704 | 1.0306 | 2.0078 | 0.9771 |
| | | $R(x)$ | 0.0439 | 0.1281 | 0.1535 | 1.0000 | 0.8510 | 0.0385 | 0.0012 | 0.1535 | 0.8767 | 0.7233 |
| | | $h(x)$ | 0.1648 | 0.0318 | 0.5041 | 1.1929 | 0.6888 | 0.1495 | 0.0041 | 0.7264 | 0.9516 | 0.2252 |
| | | MdTF | 0.1042 | 0.0585 | 1.0127 | 1.9532 | 0.9406 | 0.0957 | 0.0011 | 1.4047 | 1.5294 | 0.1247 |
| | III | $\lambda$ | 0.3622 | 0.1543 | 1.4754 | 2.9963 | 1.5209 | 0.3033 | 0.1470 | 1.3587 | 2.8377 | 1.4790 |
| | | $\theta_1$ | 0.6846 | 0.7627 | 0.4730 | 3.4184 | 2.8209 | 0.5924 | 0.2586 | 1.1230 | 2.6838 | 1.5608 |
| | | $\theta_2$ | 0.5895 | 0.5437 | 0.5974 | 3.1695 | 2.5721 | 0.5076 | 0.1722 | 1.1501 | 3.1695 | 2.0194 |
| | | $\theta_3$ | 0.8542 | 1.6144 | 0.2796 | 3.9846 | 3.7051 | 0.7258 | 0.7300 | 1.1175 | 3.1570 | 2.0395 |
| | | $\theta_4$ | 0.7748 | 1.1590 | 0.3773 | 3.7231 | 3.3458 | 0.6615 | 0.4669 | 1.1173 | 2.9524 | 1.8351 |
| | | $\theta$ | 0.3440 | 0.1937 | 0.7832 | 2.4078 | 1.6246 | 0.3221 | 0.1104 | 0.9752 | 2.1732 | 1.1980 |
| | | $R(x)$ | 0.0437 | 0.1719 | 0.0627 | 1.0000 | 0.9419 | 0.0364 | 0.0011 | 0.0627 | 0.8753 | 0.8126 |
| | | $h(x)$ | 0.1636 | 0.0300 | 0.5034 | 1.1720 | 0.6687 | 0.1472 | 0.0026 | 0.7438 | 0.9190 | 0.1752 |
| | | MdTF | 0.1037 | 0.0692 | 0.9765 | 2.0003 | 1.0238 | 0.0950 | 0.0004 | 1.4351 | 1.5135 | 0.0784 |

**Table 10.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_5$ and $M_9$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 80 | I | $\lambda$ | 0.2951 | 0.1059 | 1.5397 | 2.8045 | 1.2647 | 0.2545 | 0.1022 | 1.4494 | 2.6885 | 1.2390 |
| | | $\theta_1$ | 0.4946 | 0.3888 | 0.6821 | 2.9214 | 2.1341 | 0.4398 | 0.1295 | 1.1568 | 2.3799 | 1.2231 |
| | | $\theta_2$ | 0.4358 | 0.2841 | 0.7873 | 2.7499 | 1.9626 | 0.3877 | 0.0861 | 1.2028 | 2.7499 | 1.5471 |
| | | $\theta_3$ | 0.5298 | 0.4497 | 0.6414 | 3.0165 | 2.3751 | 0.4699 | 0.1565 | 1.1457 | 2.4633 | 1.3176 |
| | | $\theta_4$ | 0.5794 | 0.5746 | 0.5400 | 3.1859 | 2.6459 | 0.5112 | 0.2100 | 1.1102 | 2.6040 | 1.4938 |
| | | $\theta$ | 0.2725 | 0.1273 | 0.9080 | 2.2539 | 1.3459 | 0.2616 | 0.0776 | 1.0499 | 2.0866 | 1.0366 |
| | | $R(x)$ | 0.0364 | 0.1139 | 0.1846 | 1.0000 | 0.8807 | 0.0317 | 0.0008 | 0.1846 | 0.8620 | 0.6775 |
| | | $h(x)$ | 0.1307 | 0.0201 | 0.5969 | 1.1468 | 0.5499 | 0.1204 | 0.0015 | 0.8012 | 0.9327 | 0.1315 |
| | | MdTF | 0.0786 | 0.0482 | 1.0343 | 1.8909 | 0.8566 | 0.0737 | 0.0002 | 1.4230 | 1.4790 | 0.0560 |
| | II | $\lambda$ | 0.2551 | 0.0926 | 1.5300 | 2.7133 | 1.1834 | 0.2327 | 0.0906 | 1.4658 | 2.6351 | 1.1693 |
| | | $\theta_1$ | 0.4051 | 0.2803 | 0.7153 | 2.6568 | 1.8628 | 0.3761 | 0.0671 | 1.2021 | 2.0996 | 0.8975 |
| | | $\theta_2$ | 0.3768 | 0.2387 | 0.7940 | 2.6093 | 1.8152 | 0.3462 | 0.0614 | 1.2277 | 2.6093 | 1.3816 |
| | | $\theta_3$ | 0.4288 | 0.3077 | 0.6931 | 2.7244 | 2.0312 | 0.3975 | 0.0721 | 1.2085 | 2.1353 | 0.9268 |
| | | $\theta_4$ | 0.4770 | 0.3722 | 0.6297 | 2.8358 | 2.2061 | 0.4415 | 0.0832 | 1.2066 | 2.1810 | 0.9744 |
| | | $\theta$ | 0.2251 | 0.0851 | 0.9594 | 2.0729 | 1.1136 | 0.2232 | 0.0418 | 1.1070 | 1.8776 | 0.7706 |
| | | $R(x)$ | 0.0349 | 0.0771 | 0.2904 | 1.0000 | 0.7638 | 0.0312 | 0.0010 | 0.2904 | 0.8674 | 0.5769 |
| | | $h(x)$ | 0.1279 | 0.0212 | 0.5863 | 1.1526 | 0.5663 | 0.1180 | 0.0033 | 0.7584 | 0.9676 | 0.2092 |
| | | MdTF | 0.0790 | 0.0388 | 1.0809 | 1.8498 | 0.7689 | 0.0733 | 0.0009 | 1.3922 | 1.5110 | 0.1188 |
| | III | $\lambda$ | 0.2686 | 0.0993 | 1.5316 | 2.7574 | 1.2258 | 0.2378 | 0.0963 | 1.4529 | 2.6557 | 1.2028 |
| | | $\theta_1$ | 0.4482 | 0.3175 | 0.6962 | 2.7355 | 1.9567 | 0.4110 | 0.0892 | 1.1673 | 2.1852 | 1.0179 |
| | | $\theta_2$ | 0.3926 | 0.2521 | 0.7788 | 2.6281 | 1.8493 | 0.3556 | 0.0703 | 1.1954 | 2.6281 | 1.4327 |
| | | $\theta_3$ | 0.4738 | 0.3732 | 0.6701 | 2.8790 | 2.2089 | 0.4262 | 0.1104 | 1.1751 | 2.3074 | 1.1322 |
| | | $\theta_4$ | 0.5295 | 0.4822 | 0.5841 | 3.0309 | 2.4468 | 0.4761 | 0.1519 | 1.1520 | 2.4191 | 1.2671 |
| | | $\theta$ | 0.2411 | 0.1010 | 0.9224 | 2.1300 | 1.2076 | 0.2372 | 0.0563 | 1.0649 | 1.9529 | 0.8880 |
| | | $R(x)$ | 0.0352 | 0.0917 | 0.2477 | 1.0000 | 0.7556 | 0.0304 | 0.0008 | 0.2477 | 0.8658 | 0.6181 |
| | | $h(x)$ | 0.1303 | 0.0201 | 0.5845 | 1.1349 | 0.5504 | 0.1176 | 0.0022 | 0.7729 | 0.9393 | 0.1663 |
| | | MdTF | 0.0813 | 0.0442 | 1.0607 | 1.8813 | 0.8207 | 0.0738 | 0.0005 | 1.4129 | 1.4992 | 0.0864 |

**Table 11.** Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_5$ and $M_{10}$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 80 | I | $\lambda$ | 0.2966 | 0.1068 | 1.5434 | 2.8125 | 1.2691 | 0.2571 | 0.1032 | 1.4527 | 2.6986 | 1.2458 |
| | | $\theta_1$ | 0.4350 | 0.2803 | 0.7784 | 2.7195 | 2.0314 | 0.3855 | 0.0853 | 1.1897 | 2.2140 | 1.0243 |
| | | $\theta_2$ | 0.5187 | 0.4146 | 0.6880 | 2.9733 | 2.2852 | 0.4607 | 0.1426 | 1.1700 | 2.9733 | 1.8033 |
| | | $\theta_3$ | 0.4703 | 0.3526 | 0.7168 | 2.8478 | 2.1310 | 0.4164 | 0.1156 | 1.1672 | 2.3220 | 1.1548 |
| | | $\theta_4$ | 0.7335 | 0.9754 | 0.4057 | 3.6196 | 3.2139 | 0.6453 | 0.4050 | 1.1088 | 2.9972 | 1.8884 |
| | | $\theta$ | 0.2811 | 0.1284 | 0.9124 | 2.2560 | 1.3437 | 0.2696 | 0.0810 | 1.0512 | 2.1031 | 1.0519 |
| | | $R(x)$ | 0.0355 | 0.1154 | 0.1856 | 1.0000 | 0.8100 | 0.0303 | 0.0008 | 0.1856 | 0.8631 | 0.6775 |
| | | $h(x)$ | 0.1262 | 0.0200 | 0.5923 | 1.1407 | 0.5484 | 0.1144 | 0.0015 | 0.7968 | 0.9287 | 0.1318 |
| | | MdTF | 0.0760 | 0.0480 | 1.0369 | 1.8924 | 0.8555 | 0.0703 | 0.0002 | 1.4251 | 1.4805 | 0.0554 |
| | II | $\lambda$ | 0.2717 | 0.0934 | 1.5402 | 2.7280 | 1.1879 | 0.2482 | 0.0917 | 1.4762 | 2.6499 | 1.1738 |
| | | $\theta_1$ | 0.4023 | 0.2439 | 0.7933 | 2.6100 | 1.8861 | 0.3759 | 0.0637 | 1.2276 | 2.1087 | 0.8811 |
| | | $\theta_2$ | 0.4330 | 0.2914 | 0.7239 | 2.6965 | 1.9726 | 0.4007 | 0.0706 | 1.2175 | 2.6965 | 1.4790 |
| | | $\theta_3$ | 0.4114 | 0.2725 | 0.7531 | 2.6755 | 1.9224 | 0.3798 | 0.0671 | 1.2271 | 2.1317 | 0.9046 |
| | | $\theta_4$ | 0.5264 | 0.4921 | 0.5505 | 3.0551 | 2.5046 | 0.4842 | 0.1091 | 1.2222 | 2.3071 | 1.0848 |
| | | $\theta$ | 0.2327 | 0.0869 | 0.9640 | 2.0871 | 1.1231 | 0.2312 | 0.0433 | 1.1128 | 1.8953 | 0.7825 |
| | | $R(x)$ | 0.0362 | 0.0787 | 0.2869 | 1.0000 | 0.7270 | 0.0326 | 0.0009 | 0.2869 | 0.8679 | 0.5810 |
| | | $h(x)$ | 0.1296 | 0.0213 | 0.5833 | 1.1503 | 0.5669 | 0.1196 | 0.0033 | 0.7567 | 0.9657 | 0.2091 |
| | | MdTF | 0.0799 | 0.0388 | 1.0821 | 1.8505 | 0.7683 | 0.0745 | 0.0009 | 1.3940 | 1.5109 | 0.1169 |
| | III | $\lambda$ | 0.2793 | 0.0992 | 1.5281 | 2.7524 | 1.2243 | 0.2499 | 0.0962 | 1.4498 | 2.6520 | 1.2022 |
| | | $\theta_1$ | 0.3989 | 0.2604 | 0.7813 | 2.6461 | 1.9490 | 0.3607 | 0.0742 | 1.2001 | 2.1431 | 0.9430 |
| | | $\theta_2$ | 0.4451 | 0.3196 | 0.6972 | 2.7417 | 2.0445 | 0.4055 | 0.0901 | 1.1695 | 2.7417 | 1.5722 |
| | | $\theta_3$ | 0.4478 | 0.3076 | 0.7357 | 2.7615 | 2.0258 | 0.4047 | 0.0886 | 1.1942 | 2.2258 | 1.0316 |
| | | $\theta_4$ | 0.6463 | 0.7343 | 0.4735 | 3.3219 | 2.8484 | 0.5810 | 0.2548 | 1.1443 | 2.6507 | 1.5064 |
| | | $\theta$ | 0.2596 | 0.1048 | 0.9331 | 2.1566 | 1.2235 | 0.2549 | 0.0599 | 1.0716 | 1.9827 | 0.9111 |
| | | $R(x)$ | 0.0352 | 0.0945 | 0.2392 | 1.0000 | 0.7679 | 0.0311 | 0.0008 | 0.2392 | 0.8637 | 0.6245 |
| | | $h(x)$ | 0.1272 | 0.0205 | 0.5919 | 1.1486 | 0.5567 | 0.1173 | 0.0021 | 0.7838 | 0.9454 | 0.1615 |
| | | MdTF | 0.0785 | 0.0437 | 1.0566 | 1.8726 | 0.8160 | 0.0735 | 0.0005 | 1.4099 | 1.4931 | 0.0833 |

Table 12. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_6$ and $M_{11}$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 100 | I | $\lambda$ | 0.2461 | 0.0756 | 1.5849 | 2.6557 | 1.0708 | 0.2245 | 0.0740 | 1.5197 | 2.5773 | 1.0576 |
| | | $\theta_1$ | 0.3856 | 0.2324 | 0.8121 | 2.5955 | 1.7099 | 0.3543 | 0.0699 | 1.2081 | 2.1468 | 0.9387 |
| | | $\theta_2$ | 0.3600 | 0.1898 | 0.8856 | 2.5132 | 1.6276 | 0.3290 | 0.0622 | 1.2186 | 2.5132 | 1.2946 |
| | | $\theta_3$ | 0.3905 | 0.2325 | 0.8099 | 2.5910 | 1.7811 | 0.3604 | 0.0704 | 1.2054 | 2.1447 | 0.9393 |
| | | $\theta_4$ | 0.4517 | 0.3220 | 0.7138 | 2.7878 | 2.0740 | 0.4123 | 0.0888 | 1.2135 | 2.2465 | 1.0329 |
| | | $\theta$ | 0.2320 | 0.0834 | 1.0103 | 2.1091 | 1.0988 | 0.2246 | 0.0490 | 1.1307 | 1.9636 | 0.8329 |
| | | $R(x)$ | 0.0291 | 0.0754 | 0.2944 | 1.0000 | 0.7491 | 0.0268 | 0.0006 | 0.2944 | 0.8557 | 0.5613 |
| | | $h(x)$ | 0.0996 | 0.0151 | 0.6516 | 1.1298 | 0.4782 | 0.0933 | 0.0015 | 0.8195 | 0.9546 | 0.1351 |
| | | MdTF | 0.0594 | 0.0336 | 1.0901 | 1.8072 | 0.7171 | 0.0564 | 0.0003 | 1.4035 | 1.4710 | 0.0675 |
| | II | $\lambda$ | 0.2211 | 0.0697 | 1.5892 | 2.6186 | 1.0294 | 0.2051 | 0.0688 | 1.5387 | 2.5589 | 1.0202 |
| | | $\theta_1$ | 0.3459 | 0.1895 | 0.8311 | 2.4649 | 1.5583 | 0.3266 | 0.0428 | 1.2475 | 1.9927 | 0.7452 |
| | | $\theta_2$ | 0.3254 | 0.1608 | 0.9067 | 2.4133 | 1.5067 | 0.3048 | 0.0419 | 1.2601 | 2.4133 | 1.1532 |
| | | $\theta_3$ | 0.3676 | 0.2015 | 0.8445 | 2.5139 | 1.6694 | 0.3467 | 0.0468 | 1.2670 | 2.0350 | 0.7680 |
| | | $\theta_4$ | 0.3976 | 0.2602 | 0.7393 | 2.6279 | 1.8886 | 0.3742 | 0.0510 | 1.2596 | 2.0504 | 0.7908 |
| | | $\theta$ | 0.1998 | 0.0621 | 1.0453 | 2.0030 | 0.9577 | 0.1970 | 0.0306 | 1.1730 | 1.8384 | 0.6654 |
| | | $R(x)$ | 0.0302 | 0.0566 | 0.3607 | 1.0000 | 0.7166 | 0.0280 | 0.0008 | 0.3607 | 0.8607 | 0.5001 |
| | | $h(x)$ | 0.1103 | 0.0159 | 0.6395 | 1.1316 | 0.4922 | 0.1037 | 0.0028 | 0.7842 | 0.9803 | 0.1961 |
| | | MdTF | 0.0674 | 0.0285 | 1.1242 | 1.7842 | 0.6600 | 0.0636 | 0.0008 | 1.3875 | 1.4969 | 0.1095 |
| | III | $\lambda$ | 0.2362 | 0.0730 | 1.5973 | 2.6504 | 1.0530 | 0.2125 | 0.0715 | 1.5420 | 2.5815 | 1.0395 |
| | | $\theta_1$ | 0.3826 | 0.2196 | 0.8436 | 2.5925 | 1.6874 | 0.3505 | 0.0574 | 1.2600 | 2.1206 | 0.8606 |
| | | $\theta_2$ | 0.3487 | 0.1775 | 0.9051 | 2.4775 | 1.5724 | 0.3223 | 0.0522 | 1.2531 | 2.4775 | 1.2244 |
| | | $\theta_3$ | 0.3698 | 0.2151 | 0.8354 | 2.5637 | 1.7282 | 0.3428 | 0.0563 | 1.2483 | 2.0966 | 0.8482 |
| | | $\theta_4$ | 0.4208 | 0.2854 | 0.7274 | 2.6850 | 1.9576 | 0.3922 | 0.0664 | 1.2380 | 2.1260 | 0.8880 |
| | | $\theta$ | 0.2069 | 0.0717 | 1.0378 | 2.0641 | 1.0263 | 0.2011 | 0.0384 | 1.1635 | 1.9070 | 0.7434 |
| | | $R(x)$ | 0.0302 | 0.0650 | 0.3291 | 1.0000 | 0.7811 | 0.0276 | 0.0007 | 0.3291 | 0.8592 | 0.5301 |
| | | $h(x)$ | 0.1068 | 0.0155 | 0.6454 | 1.1312 | 0.4858 | 0.1013 | 0.0022 | 0.7993 | 0.9704 | 0.1711 |
| | | MdTF | 0.0637 | 0.0305 | 1.1092 | 1.7924 | 0.6832 | 0.0610 | 0.0005 | 1.3936 | 1.4843 | 0.0907 |

Table 13. Point and interval estimates on $R(x=1), h(x=1)$ and MdTF for $N_6$ and $M_{12}$.

| | | | ML Estimates | | | | | Pivotal Estimates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACIs | | | | | GCIs | | |
| $n$ | CS | Pars | Bias | Variance | Lower | Upper | Lengths | Bias | Variance | Lower | Upper | Lengths |
| 100 | I | $\lambda$ | 0.2523 | 0.0812 | 1.5692 | 2.6788 | 1.1096 | 0.2286 | 0.0793 | 1.4972 | 2.5914 | 1.0942 |
| | | $\theta_1$ | 0.4322 | 0.2865 | 0.7612 | 2.7336 | 1.9206 | 0.3926 | 0.0940 | 1.1825 | 2.2672 | 1.0846 |
| | | $\theta_2$ | 0.3872 | 0.2333 | 0.8130 | 2.5953 | 1.7823 | 0.3529 | 0.0836 | 1.1654 | 2.5953 | 1.4300 |
| | | $\theta_3$ | 0.3931 | 0.2359 | 0.8020 | 2.6010 | 1.7991 | 0.3590 | 0.0735 | 1.1880 | 2.1543 | 0.9662 |
| | | $\theta_4$ | 0.5174 | 0.4048 | 0.6736 | 2.9282 | 2.2546 | 0.4771 | 0.1256 | 1.1976 | 2.3709 | 1.1733 |
| | | $\theta$ | 0.2534 | 0.0979 | 0.9700 | 2.1544 | 1.1845 | 0.2441 | 0.0598 | 1.0965 | 2.0100 | 0.9134 |
| | | $R(x)$ | 0.0293 | 0.0876 | 0.2562 | 1.0000 | 0.7269 | 0.0264 | 0.0006 | 0.2562 | 0.8550 | 0.5988 |
| | | $h(x)$ | 0.1079 | 0.0160 | 0.6420 | 1.1346 | 0.4926 | 0.0999 | 0.0011 | 0.8264 | 0.9410 | 0.1146 |
| | | MdTF | 0.0649 | 0.0381 | 1.0704 | 1.8327 | 0.7623 | 0.0607 | 0.0002 | 1.4153 | 1.4674 | 0.0520 |
| | II | $\lambda$ | 0.2265 | 0.0734 | 1.5815 | 2.6370 | 1.0555 | 0.2063 | 0.0718 | 1.5303 | 2.5731 | 1.0428 |
| | | $\theta_1$ | 0.3805 | 0.2253 | 0.8059 | 2.5705 | 1.7103 | 0.3570 | 0.0513 | 1.2568 | 2.0588 | 0.8020 |
| | | $\theta_2$ | 0.3314 | 0.1795 | 0.8601 | 2.4547 | 1.5945 | 0.3086 | 0.0447 | 1.2415 | 2.4547 | 1.2132 |
| | | $\theta_3$ | 0.3519 | 0.1992 | 0.8362 | 2.5023 | 1.6662 | 0.3322 | 0.0471 | 1.2530 | 2.0291 | 0.7761 |
| | | $\theta_4$ | 0.4197 | 0.2879 | 0.7122 | 2.6846 | 1.9724 | 0.3952 | 0.0569 | 1.2569 | 2.0796 | 0.8227 |
| | | $\theta$ | 0.1985 | 0.0670 | 1.0276 | 2.0221 | 0.9945 | 0.1974 | 0.0327 | 1.1621 | 1.8497 | 0.6876 |
| | | $R(x)$ | 0.0306 | 0.0608 | 0.3447 | 1.0000 | 0.7050 | 0.0278 | 0.0008 | 0.3447 | 0.8619 | 0.5173 |
| | | $h(x)$ | 0.1087 | 0.0170 | 0.6279 | 1.1366 | 0.5087 | 0.1011 | 0.0028 | 0.7802 | 0.9760 | 0.1958 |
| | | MdTF | 0.0663 | 0.0303 | 1.1153 | 1.7953 | 0.6800 | 0.0619 | 0.0008 | 1.3890 | 1.4976 | 0.1086 |
| | III | $\lambda$ | 0.2495 | 0.0789 | 1.5918 | 2.6859 | 1.0941 | 0.2212 | 0.0771 | 1.5298 | 2.6087 | 1.0789 |
| | | $\theta_1$ | 0.4156 | 0.2609 | 0.7915 | 2.6695 | 1.8198 | 0.3808 | 0.0734 | 1.2283 | 2.1770 | 0.9487 |
| | | $\theta_2$ | 0.3736 | 0.2144 | 0.8498 | 2.5722 | 1.7224 | 0.3431 | 0.0669 | 1.2200 | 2.5722 | 1.3521 |
| | | $\theta_3$ | 0.3831 | 0.2232 | 0.8280 | 2.5714 | 1.7433 | 0.3550 | 0.0617 | 1.2321 | 2.1060 | 0.8739 |
| | | $\theta_4$ | 0.4529 | 0.3251 | 0.6870 | 2.7675 | 2.0805 | 0.4173 | 0.0816 | 1.2107 | 2.1946 | 0.9840 |
| | | $\theta$ | 0.2306 | 0.0825 | 1.0052 | 2.0997 | 1.0945 | 0.2246 | 0.0457 | 1.1344 | 1.9396 | 0.8053 |
| | | $R(x)$ | 0.0320 | 0.0746 | 0.2998 | 1.0000 | 0.7451 | 0.0284 | 0.0007 | 0.2998 | 0.8609 | 0.5612 |
| | | $h(x)$ | 0.1145 | 0.0162 | 0.6304 | 1.1260 | 0.4956 | 0.1067 | 0.0019 | 0.7949 | 0.9525 | 0.1576 |
| | | MdTF | 0.0688 | 0.0342 | 1.0962 | 1.8188 | 0.7227 | 0.0649 | 0.0004 | 1.4061 | 1.4863 | 0.0802 |

Table 14. Fatalities in coal mining disasters in Britain.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1900–1919 | 0.20 | 0.22 | 0.24 | 0.26 | 0.28 | 0.32 | 0.50 | 0.54 |
| | 0.54 | 0.66 | 0.70 | 0.78 | 1.04 | 1.50 | 1.62 | 1.76 |
| | 2.38 | 2.72 | 3.36 | 6.88 | 8.78 | | | |
| 1920–1939 | 0.20 | 0.20 | 0.22 | 0.22 | 0.24 | 0.26 | 0.26 | 0.28 |
| | 0.28 | 0.38 | 0.54 | 0.54 | 0.54 | 0.60 | 0.78 | 0.90 |
| | 1.04 | 1.16 | 1.58 | 5.30 | | | | |
| 1940–1959 | 0.20 | 0.22 | 0.24 | 0.24 | 0.24 | 0.26 | 0.30 | 0.30 |
| | 0.32 | 0.34 | 0.38 | 0.42 | 0.44 | 0.90 | 1.14 | 1.66 |
| | 2.08 | | | | | | | |

Table 15. K–S statistics and the *p*-values of IER distribution for real data.

| Facilities | MLEs | K–S distance | *p*-value |
|---|---|---|---|
| 1900–1919 | $\hat{\lambda} = 0.2666685, \hat{\theta} = 0.42782761$ | 0.1239 | 0.9038 |
| 1920–1939 | $\hat{\lambda} = 0.2377864, \hat{\theta} = 0.72198203$ | 0.1944 | 0.4363 |
| 1940–1959 | $\hat{\lambda} = 0.2658330, \hat{\theta} = 0.96312893$ | 0.1685 | 0.7200 |

considered as the different facilities. (Hand et al., 1993) reported the number of fatalities from year 1840 to 1959 with a period of 20 years. In this analysis, we have considered last three periods as three different facilities from 1900 to 1959 and reported the data in Table 14. For the ease of the computation, we divide each observation in the data by 30.

Before analyzing this data set, we first perform goodness-of-fit for the selected IER distribution. In this regard, we obtained the maximum likelihood estimates and Kolmogorov–Smirnov test statistics along with corresponding p-values as reported in Table 15. From the table, it can be concluded that the IER distribution may considered as an appropriate model to analyze this data set. Further, we have plotted empirical cdf overlaying with theoretical cdf, p–P plot and Q–Q plot, which also showed the significant goodness-of-fit for the IER distribution as shown in Figure 1.

Based on the complete data shown in Table 15, a block progressive censored data is generated where three periods are considered to be as different test facilities. Specifically, the generated block progressive censored data from three different periods have sample sizes $(n_1, n_2, n_3) = (21, 20, 17)$ and effective sample sizes $(m_1, m_2, m_3) = (18, 17, 15)$, and the same progressive censoring schemes are taken in each group to be $R = (0, \ldots, 0, n_i - m_i)$ being traditional Type-II censoring scenario, and the block progressive censored data for fatalities in coal mining disasters of each period are presented in Table 16.

Now, in order to model the number of fatalities in coal mining disasters, we use IER$(\theta_i, \lambda)$ distribution with the generated block progressive censored data given in Table 16. The estimate of $\theta_i$ will take in to account the study of differences in number of fatalities during $i$th period (i.e. difference in different test facilities). Under the generated data set, point estimates along with asymptotic confidence intervals (ACIs) and generalized confidence intervals (GCIs) are obtained using maximum likelihood and pivotal quantity-based approaches. In order to achieve pivotal-based estimates, the estimated quantities are generated 2000 times. Model parameters as well as survival characteristics $R(1)$, $h(1)$ and MdTF are estimated under classical likelihood and pivotal quantity method, and the associated results are presented in Table 17. The interval estimates are obtained at 95% significance level. The estimated results of $\theta_i$ can be used to study the differences in number of fatalities during different periods.

From Table 17, it is observed that the estimates obtained using pivotal quantity approach are having smaller variances and interval lengths as compared to maximum likelihood approach. This indicated that the pivotal quantity approach is superior than the traditional maximum likelihood approach. Moreover, the interval estimates for $\theta_1, \theta_2, \theta_3$ exhibit considerable overlap that further indicates that the three time periods are consistent in our case. Therefore, it is vitally necessary to take category consistency into account showing that the inferences will be severely affected if this

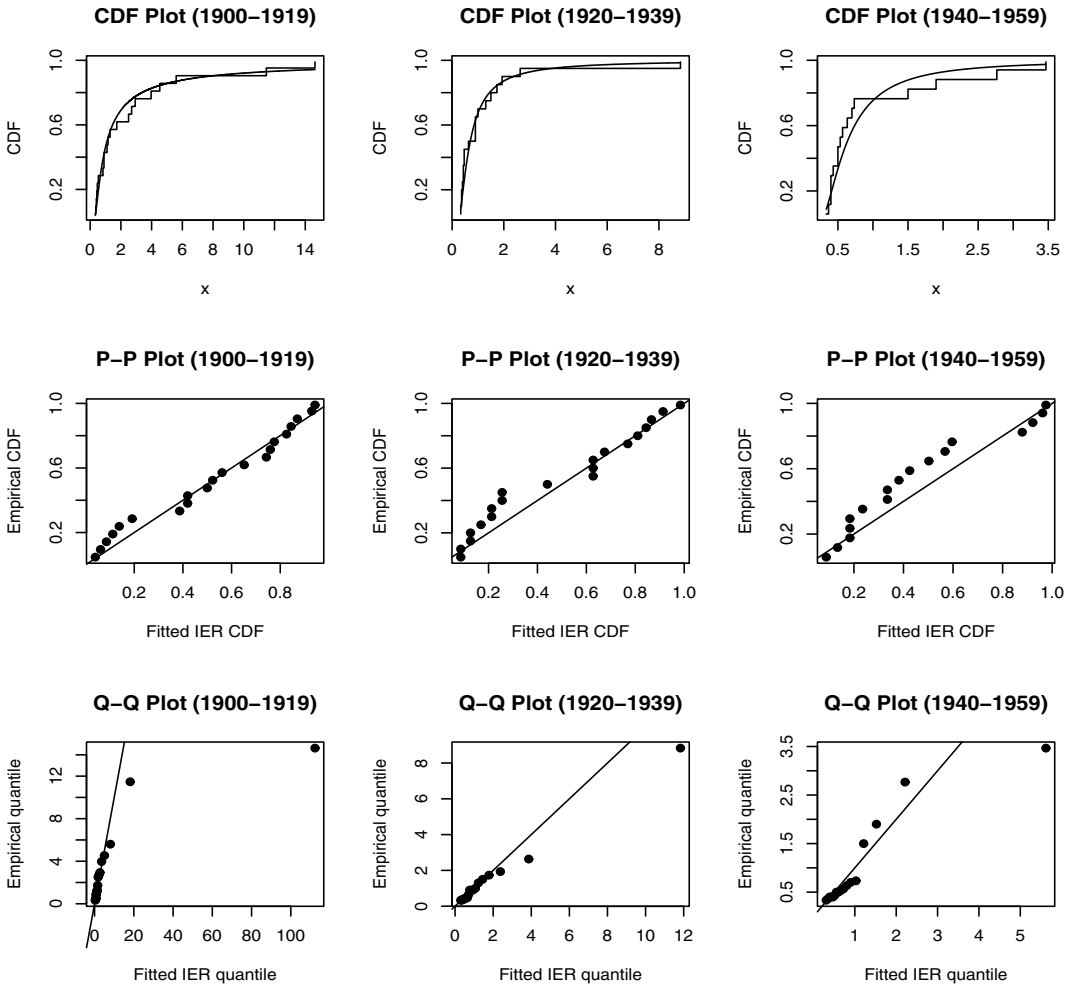



Figure 1. Plot for uniqueness of MLE under real data set.

Table 16. Generated block progressive censored data for fatalities in coal mining disasters.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1900–1919 | 0.20 | 0.22 | 0.24 | 0.26 | 0.28 | 0.32 | 0.50 | 0.54 |
| | 0.54 | 0.66 | 0.70 | 0.78 | 1.04 | 1.50 | 1.62 | 1.76 |
| | 2.38 | 2.72 | | | | | | |
| 1920–1939 | 0.20 | 0.20 | 0.22 | 0.22 | 0.24 | 0.26 | 0.26 | 0.28 |
| | 0.28 | 0.38 | 0.54 | 0.54 | 0.54 | 0.60 | 0.78 | 0.90 |
| | 1.04 | | | | | | | |
| 1940–1959 | 0.20 | 0.22 | 0.24 | 0.24 | 0.24 | 0.26 | 0.30 | 0.30 |
| | 0.32 | 0.34 | 0.38 | 0.42 | 0.44 | 0.90 | 1.14 | |

consistency is disregarded and each category is thought to be the product of a independent source. Furthermore, since there is incomplete overlap in the 95% intervals estimates, it is also evident that the intervals estimates for $\theta_1, \theta_2, \theta_3$ exhibit some differences in number of fatalities across three periods. This observation is consistent with the findings in Section 4. To sum up, this real-life application example demonstrates once more the superiority of the suggested pivotal quantity method over the classical likelihood results.

Table 17. Point and interval estimates on $R(x=1), h(x=1)$, MdTF and the DIDTF for fatalities data in coal mining disasters.

| | | | ACIs | | | | | GCIs | | |
|---|---|---|---|---|---|---|---|---|---|---|
| parameter | MLE | Variance | Lower | Upper | Lengths | Pivotal Quantity | Variance | Lower | Upper | Lengths |
| $\theta_1$ | 0.3954 | 0.0095 | 0.2040 | 0.5869 | 0.3829 | 0.3791 | 0.0010 | 0.3190 | 0.4376 | 0.1186 |
| $\theta_2$ | 0.7474 | 0.0407 | 0.3518 | 1.1430 | 0.7912 | 0.7015 | 0.0082 | 0.5330 | 0.8810 | 0.3480 |
| $\theta_3$ | 0.9159 | 0.0707 | 0.3946 | 1.4371 | 0.9425 | 0.8542 | 0.0151 | 0.6282 | 1.1005 | 0.4723 |
| $\theta$ | 0.5069 | 0.0085 | 0.3263 | 0.6875 | 0.3612 | 0.5008 | 0.0030 | 0.3975 | 0.6097 | 0.2122 |
| $\lambda$ | 0.2506 | 0.0025 | 0.1524 | 0.3487 | 0.1963 | 0.2258 | 0.0012 | 0.1352 | 0.3275 | 0.1023 |
| $R(x)$ | 0.4659 | 0.0054 | 0.3213 | 0.6106 | 0.2893 | 0.4483 | 0.0001 | 0.4370 | 0.4685 | 0.0315 |
| $h(x)$ | 0.8921 | 0.5180 | 0.5185 | 2.3027 | 1.7847 | 0.8905 | 0.0056 | 0.7407 | 1.0275 | 0.2868 |
| MdTF | 0.9231 | 0.0393 | 0.5347 | 1.3114 | 0.7767 | 0.8815 | 0.0007 | 0.8335 | 0.9311 | 0.0976 |

Table 18. Generated block progressive censored life test data for three operators on a component in cutting tool.

| Operator Number | Fifty Thousands of Cycles | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Op1 | 0.12 | 0.30 | 0.34 | 0.42 | 0.42 | 0.44 | 0.44 | 0.46 |
| | 0.46 | 0.52 | 0.56 | 0.56 | 0.60 | 0.78 | 0.82 | 0.94 |
| | 1.00 | 1.08 | | | | | | |
| Op2 | 0.36 | 0.50 | 0.52 | 0.60 | 0.64 | 0.68 | 0.76 | 0.76 |
| | 0.82 | 0.88 | 1.00 | 1.28 | 1.36 | 1.54 | 1.66 | 1.68 |
| | 1.74 | | | | | | | |
| Op3 | 0.44 | 0.80 | 0.92 | 1.46 | 1.74 | 2.28 | 2.54 | 2.58 |
| | 2.66 | 2.86 | 3.22 | 3.62 | 4.28 | 4.34 | 4.78 | 4.80 |
| | 5.26 | 5.44 | | | | | | |

## 5.2. Real data 2

The were from an experiment to compare three operators who performed life tests on a component in a cutting tool and is used in (Meeker et al., 2022), Sections 12.2 and 12.5. Here, three operators are considered as three test facilities to study the test facility effect. For the ease of computation, observations in this data set are divided by 50. We have also checked the fitting of proposed distribution for this data set as well using the same test statistic and plots as for Real Data 1 but not reported the results for the spacious convenience. From the results, the proposed distribution also showed significant fitting to the Real Data 2. Then, a block progressive censored data from the complete data is generated and reported in Table 18 where three operators are considered to be as different test facilities. The generated data set have samples sizes $(n_1, n_2, n_3) = (20, 20, 20)$ and effective sample sizes $(m_1, m_2, m_3) = (18, 17, 18)$, and the same progressive censoring schemes are taken in each group to be $R = (0, \ldots, 0, n_i - m_i)$ being traditional Type-II censoring scenario.

Furthermore, we model the block progressive censored data provided in Table 18 using the IER$(\theta_i, \lambda)$ distribution. The estimation of $\theta_i$ takes into account difference in life-testing specific to the $i$th operator, representing differences in different test facilities. In the context of the generated dataset, point estimates for all quantities of interest, along with ACIs and GCIs, are computed and presented in Table 19. The pivotal sample size, mission time and significance level for these estimates are similar as chosen for Real Data 1. The estimated results of $\theta_i$ can be used to study the differences in life tests for different operators. From Table 19, results indicate that the pivotal quantity approach is superior than the traditional maximum likelihood approach for this data set as well. The estimates of $\theta_i$'s found to be significantly different from each other. Moreover, the interval estimates for $\theta_1, \theta_2, \theta_3$ exhibit incomplete overlap. Hence, it indicates that there are differences in life testing across three operators. This observation is consistent with the findings in Section 4. This real-life application example also demonstrates once more the superiority of the suggested pivotal quantity method over the classical likelihood results and the DDTF cannot be ignored.

Table 19. Point and interval estimates on $R(x=1), h(x=1)$, MdTF and the DIDTF for life test data for three operators on a component in cutting tool.

| | | | ACIs | | | | | GCIs | | |
|---|---|---|---|---|---|---|---|---|---|---|
| parameter | MLE | Variance | Lower | Upper | Lengths | Pivotal Quantity | Variance | Lower | Upper | Lengths |
| $\theta_1$ | 1.0623 | 0.0953 | 0.4572 | 1.6674 | 1.2102 | 1.3552 | 0.0453 | 0.9631 | 1.7902 | 0.8271 |
| $\theta_2$ | 0.5000 | 0.0178 | 0.2388 | 0.7611 | 0.5223 | 0.5829 | 0.0034 | 0.4690 | 0.6976 | 0.2286 |
| $\theta_3$ | 0.2500 | 0.0037 | 0.1310 | 0.3690 | 0.2380 | 0.2704 | 0.0002 | 0.2316 | 0.2966 | 0.0650 |
| $\theta$ | 0.3169 | 0.0034 | 0.2024 | 0.4313 | 0.2289 | 0.3550 | 0.0009 | 0.2972 | 0.4082 | 0.1110 |
| $\lambda$ | 0.2076 | 0.0021 | 0.1174 | 0.2979 | 0.1805 | 0.2788 | 0.0025 | 0.1821 | 0.3791 | 0.1970 |
| $R(x)$ | 0.5883 | 0.0037 | 0.4686 | 0.7080 | 0.2394 | 0.6047 | 0.0002 | 0.5726 | 0.6280 | 0.0554 |
| $h(x)$ | 0.5702 | 0.3075 | 0.0000 | 1.6571 | 1.6571 | 0.6146 | 0.0016 | 0.5421 | 0.6719 | 0.1298 |
| MdTF | 1.3208 | 0.1391 | 0.5897 | 2.0519 | 1.4622 | 1.3479 | 0.0015 | 1.2102 | 1.4407 | 0.2305 |

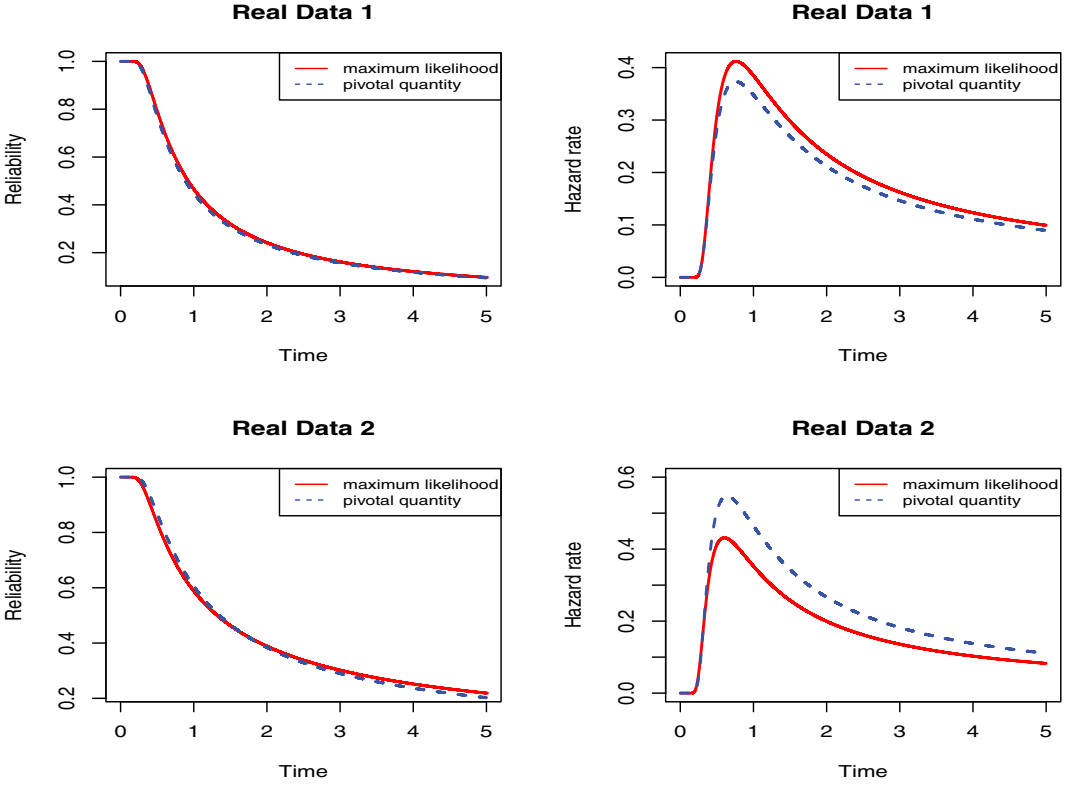



Figure 2. Changing tendencies of reliability and hazard rate along with time by using maximum likelihood and pivotal quantity for real data 1 and 2.

Finally, based on the results presented in Tables 17 and 19, another plots are shown in Figure 2 to illustrate how the reliability and hazard rate function patterns have changed over different mission time using the parameter estimates obtained by maximum likelihood and pivotal quantity estimation, respectively. In Figure 2, it is noted that reliability function significantly declines with time while hazard rate function has uni-model shape in consequence with respect to both datasets. In addition, graphs of the profile log-likelihood function of the parameter $\lambda$ for both datasets are also lotted in Figure 3, and the graphs indicate the existence and uniqueness of the MLE of $\lambda$ that has consistency with Theorem 2.2.

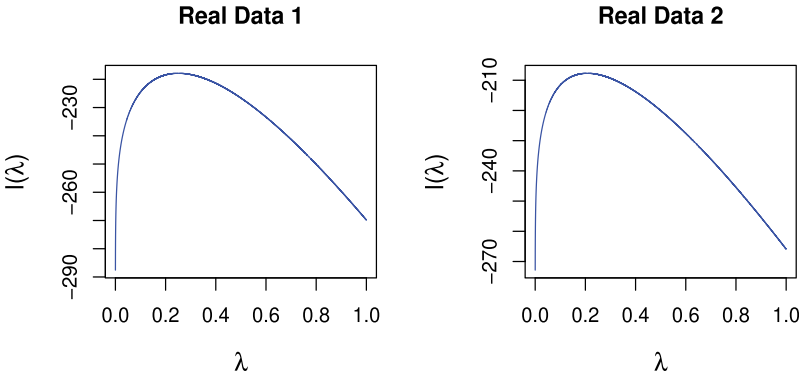



Figure 3. Profile log-likelihood plots for uniqueness of MLEs under real data 1 and 2.

## 6. Conclusions

This paper has undertaken a comprehensive analysis of survival characteristics, particularly focusing on the reliability function, hazard rate function, median time to failure, and differences in different test facilities. The study employed block progressive censored data and considered an inverse exponentiated Rayleigh distribution as the underlying lifetime distribution for test units. Maximum likelihood and pivotal methods were utilized for parameter estimation, and the paper-derived estimators while exploring their properties. The simulation study demonstrated that the pivotal quantity-based approach outperformed other methods in terms of estimation accuracy. The proposed estimators were then applied to analyze two real datasets, showcasing their practical applicability. The results revealed notable consistency between test facilities as indicated by considerable overlap in interval estimates. However, incomplete overlapping was also observed which indicated differences among test facilities. Moreover, the behavior of reliability and hazard rate estimates was depicted through plots, highlighting the significant decline of the reliability function with time and the uni-modal shape of the hazard rate function for both datasets. The research findings have significant managerial implications for organizations in testing and quality assurance. The identification of consistency and differences in parameter estimates across test facilities offers valuable insights into the reliability and performance of tested units. This information allows managers to assess product uniformity across different testing environments, aiding in decision-making related to product performance. The recommendation to adopt the pivotal quantity-based approach for enhanced accuracy in reliability analysis highlights the importance of implementing robust estimation methods. Extending the challenge of inference for various lifetime distributions involves the application of block censored mechanisms within alternative censoring methods, including hybrid censoring, adaptive progressive hybrid censoring, generalized progressive hybrid censoring, progressive interval censoring methods, among others. Exploring optimal designs for block censored life-testing experiments could be a potential work for future research. This entails enhancing our understanding of the implications and advantages of employing block censored mechanisms in diverse censoring scenarios.

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No potential conflict of interest was reported by the author(s).

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